

Measuring Scientific Decision-Making: Development and Validation

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Abstract

Scientific decision-making (SDM), defined as the structured use of causal reasoning, evidence gathering and evaluation, and adaptive learning to navigate uncertainty, is theoretically central to management, yet no validated self-report instrument exists for field-scale deployment. We develop and validate a 24-item SDM instrument, specified as a formative construct, with three dimensions, and apply it to a field sample of 9,458 managers from 61 firms. Validation accumulates across five complementary criteria: content validity and expert review, indicator non-redundancy (VIF: 1.03–1.19), experimental sensitivity, structural distinguishability of subdimensions, and predictive validity via vignette decision accuracy. In a controlled experiment with practicing managers ($N = 462$), scientific priming raised SDM scores compared to placebo ($d = 1.40$) and intuitive priming ($d = 2.09$), confirming causal sensitivity. In the field sample, SDM follows the organizational hierarchy: CEO and senior managers score progressively higher than frontline managers, and this monotonic gradient survives firm fixed effects. The gradient is not uniform across dimensions: Adaptive Learning differentiates tiers ($\Delta = .19$ scale points) while Evidence Evaluation does not ($\Delta = .04$), indicating that hierarchy structures *how* managers reason, not *whether* they seek evidence. SDM scores are associated with vignette performance: top-quintile managers give responses consistent with scientific decision-making at rates 12 percentage points higher than bottom-quintile managers. The instrument makes scientific reasoning observable at scale, opening up cumulative research on the antecedents, contingencies, and consequences of SDM in organizations.

Keywords: scientific decision-making, measurement, scale development, construct validity, formative measurement, cumulative evidence, experimental validation, vignette validation

1 Introduction

Scientific decision-making (SDM) addresses a fundamental challenge: managers operate under uncertainty about states, actions, and consequences that cannot be resolved by optimization alone (Camuffo et al., 2020a, 2024b). Randomized trials across multiple industries and national contexts confirm that SDM training improves decision quality and learning, suggesting that the construct captures a generalizable logic of managerial reasoning rather than a context-specific skill (Camuffo et al., 2024b; Zellweger and Zenger, 2023). However, the field lacks an instrument to assess whether and how managers already practice SDM outside experimental settings.

This absence is a bottleneck at the field level. Existing studies adopt a scientific approach through intervention; they do not measure the extent to which managers already practice it. This design precludes large-sample observational research, cross-organizational comparison, and the study of SDM as an antecedent, consequence, or moderator of other constructs. Without a validated instrument, the field cannot describe how SDM is distributed across organizations, whether it follows predictable within-firm patterns, or how it relates to decision quality in natural settings. Cumulative empirical science about SDM requires a measurement infrastructure that does not yet exist.

We address this gap with two contributions. First, we develop and validate a 24-item survey instrument that operationalizes SDM at the individual level. SDM is specified as a formative construct: the three components (Causal Reasoning, Evidence Evaluation, and Adaptive Learning) define it rather than reflecting a single latent trait (Jarvis et al., 2003; Diamantopoulos and Winklhofer, 2001). Validation evidence accumulates across five complementary criteria: content validity, indicator non-redundancy, experimental sensitivity, structural distinguishability of subdimensions, and behavioral prediction. Second, we deploy the instrument in a field sample of 9,458 managers across 61 firms, documenting the distribution of SDM within and between organizations. Together, the two contributions transform SDM from an experimentally induced treatment into an observable and comparable characteristic of managers.

The instrument deployment reveals three substantive patterns. SDM follows the organi-

zational hierarchy: CEOs and senior managers score consistently higher than frontline managers, and this gradient holds within firms after controlling for composition. This finding is theoretically consequential: if the organizational position shapes *how* managers reason and not merely their access to resources or information, then the hierarchy may function as a cognitive stratification mechanism — though the cross-sectional design cannot establish causal direction. The gradient is not uniform across dimensions: Adaptive Learning differentiates hierarchical tiers more sharply than Evidence Evaluation, indicating that the organizational level shapes the *approach* to reasoning, not just its frequency. SDM scores are also associated with vignette performance, providing initial evidence that the instrument captures variance in decision behavior rather than self-reported orientation alone.

This paper does not explain why SDM levels differ across organizations or hierarchical positions; those questions require different designs and are deferred to future work. Moreover, it does not test performance consequences at the firm level. The instrument is the contribution: it makes scientific reasoning observable at scale and creates the measurement infrastructure for cumulative research on the antecedents, contingencies, and consequences of SDM in organizations.

Section 2 develops the theoretical foundations and motivates the three-dimensional formative structure. Instrument development is described in Section 3. Section 4 presents the cumulative validation evidence. Section 5 deploys the instrument in the field sample, documenting the hierarchical gradient and predictive validity. Limitations and implications follow.

2 Theoretical Framework

2.1 Scientific Decision-Making as a Structured Process

The scientific approach to managerial decision-making refers to managers adopting a protocol that mirrors scientific inquiry when making decisions under uncertainty (Camuffo et al., 2020a, 2024b; Zellweger and Zenger, 2023). Managers face fundamental uncertainty about states, actions, and consequences. Scientific decision-making (SDM) captures how managers structure this uncertainty

through theorizing, systematic evidence generation and analysis, and belief revision.

SDM is conceptualized as a process-based approach rather than a stable trait or preference. The process unfolds through six analytically distinct steps: problem diagnosis, causal theorizing, hypothesis formulation, evidence collection, evidence analysis and evaluation, and belief adjustment. The logic is sequential; the practice is not. Managers go through these steps, refining their understanding as new information emerges (Camuffo et al., 2020a, 2024b).

2.2 The Core Components of Scientific Decision-Making

The six steps are clustered into three higher-order cognitive processes that define SDM. Building on foundational work on causal inference (Pearl and Mackenzie, 2018), recent research has articulated how entrepreneurs engage in sequential processes of belief formation, testing, and response (Zellweger and Zenger, 2023). Each component is necessary; omitting any would fundamentally alter the construct.

Causal Reasoning Managers construct theories that causally connect hypothetical conditions, actions, and future events. A theory creates a future state space, a set of possibilities the manager believes plausible and valuable. Theories clarify which dimensions are received and which are neglected (Felin and Zenger, 2009, 2017a; Ehrig and Schmidt, 2022). Given resource constraints, managers who formulate theories develop multiple alternative theories before they commit (Camuffo et al., 2024c). This reflects awareness that not only there is uncertainty about how actions map onto consequences, but the choice of actions, states, and outcomes is itself uncertain.

The causal structure of theories enables the decomposition of managerial problems into testable components (Camuffo et al., 2024b). Hypotheses specify the causal reasoning underlying the theory and represent testable and falsifiable statements about states, actions, or outcomes. These hypotheses follow logically from the theory and lay the groundwork for systematic testing.

Evidence Evaluation Managers design and conduct experiments to collect additional information about theorized relationships. Experiments are disciplined efforts to test beliefs about specific components of a theory or about the theory itself. Well-designed experiments allow managers to collect informative data on the hypothesized relationships (Blackwell, 1953; Kerr et al., 2014; Koning et al., 2022).

Managers adopting the scientific approach design experiments to maximize learning. The choice of experimental design involves trade-offs in the type of information prioritized (Gans, 2023). The quality of evidence collection and evaluation determines the informativeness of the feedback loop.

Adaptive Learning Managers use experimental evidence to update their beliefs: the perceived likelihoods of theorized states coming to fruition (Camuffo et al., 2024c). These subjective beliefs represent the extent to which managers are persuaded or doubtful that their theory will lead to the desired outcome. Managers adopting the scientific approach recognize that their perceptions may be biased or wrong. They gather information to reshape their preconceptions and reassess the plausibility of their theories.

Belief updating leads managers either to revise decisions about outcomes given their theory or to question the theory itself, making decisions about their assumptions. This adaptive learning is different from evidence evaluation. High-quality evidence-gathering does not ensure belief updating; managers may rigorously test certain rigid beliefs (Tripsas and Gavetti, 2000a).

2.3 Measurement Implications

Building on this theoretical framework, we operationalize SDM through a survey instrument designed to measure how scientifically managers approach decision-making under uncertainty. The instrument captures managers' orientation toward each step of the scientific process across the three higher-order dimensions.

SDM is specified as a formative construct. Its indicators define the construct rather than

reflecting a single latent trait. The distinction matters for measurement. In a reflective model—the default in management research—items are effects of a common latent cause: each is a reflection of the same underlying construct, and any item can in principle substitute for another without loss of validity (Jarvis et al., 2003; Diamantopoulos and Winklhofer, 2001). Job satisfaction items, for example, co-vary because satisfaction causes each response; removing one item leaves the construct intact. SDM inverts this logic. Its items are causes of the construct, not effects. Causal reasoning, evidence evaluation, and adaptive learning together constitute scientific decision-making; no dimension substitutes for another. A manager who excels at causal reasoning but does not update beliefs based on evidence is not practicing SDM—one component’s absence changes what is being measured. Low inter-item correlations within and across dimensions are therefore expected and appropriate: they reflect genuine non-redundancy, not poor measurement.

The instrument provides a general measure of how scientifically managers approach decision-making and allows investigation of how SDM is related to decision quality and organizational performance.

3 Instrument Development

The SDM survey was designed to operationalize the theoretically specified construct described in Section 2. The instrument was built deductively: items were derived from the conceptual domain established in the literature and refined through successive stages of face, content, construct, and criterion validity checks.

3.1 Conceptual Domain and Item Generation

Section 2 established that SDM comprises six analytically distinct processes—problem definition, causal theorizing, hypothesis formulation, evidence gathering, evidence evaluation, and belief adjustment, clustered into three higher-order dimensions: causal reasoning (steps 1–3), evidence evaluation (steps 4–5), and adaptive learning (step 6) (Camuffo et al., 2020a, 2024b; Zellweger

and Zenger, 2023). This dimensional structure served as an explicit blueprint for item generation.

Table 1: Conceptual mapping of SDM processes to theoretical foundations

SDM Factor	SDM Steps	Theoretical Foundations	Cognitive Process	Role in Scientific Method	Formative Necessity	Loss If Omitted
Causal Reasoning	Steps 1–3: Problem framing; theory development; hypothesis creation	Theory-based view (Felin and Zenger, 2017b); pragmatist epistemology (Peirce, 1878); sense-making (Weick, 1995)	Construct causal models; articulate theorized relationships among problem dimensions; formulate explicit hypotheses about value creation mechanisms	Belief formation: translate strategic problem into testable propositions; establish prior theoretical commitments against which evidence will be evaluated	Foundational. Without structured theory, evidence gathering becomes ad hoc sampling; managers lack an organizing framework for hypothesis testing. Enables falsifiable predictions (Popper, 1934).	Managers collect reactive data without a coherent problem model; evidence becomes disconnected from strategic logic; indiscriminate pivoting without learning.
Evidence Evaluation	Steps 4–5: Designing tests; collecting, analyzing, and reporting information	Hypothesis testing theory (Klayman and Ha, 1987); behavioral decision research (Stanovich and Toplak, 2023); inquiry as distinct from belief (Dewey, 1938)	Systematically collect, appraise, and weigh evidence against prior hypotheses; distinguish confirmatory from disconfirmatory signals; assess evidence quality and relevance	Empirical assessment: design and conduct rigorous tests of causal propositions; gather diagnostic information sufficient to discriminate competing explanations	Essential complement to theorizing. Theorizing without rigorous evidence gathering devolves to armchair speculation. Evidence evaluation rigor determines the informativeness of the feedback loop (Klahr and Dunbar, 1988).	Managers rely on intuitive judgment, selective anecdotes, or availability heuristics; fail to generate disconfirmatory evidence; miss critical signals due to poor experimental design.
Adaptive Learning	Step 6: Deciding—acting or revising the theory	Belief updating literature (Edwards, 1968; Rollwage et al., 2020); conceptual change theory (Chi, 2008); organizational learning (Cohen and Levinthal, 1990)	Revise beliefs, goals, or strategies proportionally to the strength and direction of evidence; abandon or modify prior theories when disconfirming evidence accumulates; maintain epistemic flexibility despite identity or sunk-cost constraints	Belief revision: integrate evidence into revised mental models; implement strategic adjustments aligned with updated causal understanding; break commitment to falsified theories	Distinct from evidence evaluation. Confirmation bias research shows that high-quality evidence gathering does <i>not</i> ensure belief updating (Rollwage et al., 2020). Managers may test rigorously yet maintain rigid beliefs (Polaroid case; Tripsas and Gavetti 2000b).	Managers gather evidence effectively but dismiss contradictory findings; escalate commitment to failing strategies; experience slow or incomplete organizational learning despite access to valid information.

Note. The table maps the three higher-order SDM factors to their constituent theoretical literatures, cognitive processes, and roles in the scientific reasoning sequence. SDM is specified as a formative construct: each factor is a necessary and non-interchangeable component of scientific decision-making, and the three factors are not expected to be highly intercorrelated.

Items were generated deductively from the framework and grounded in semi-structured interviews with 12 senior managers.¹ Interviewees described recent low-frequency, high-impact decisions made under high uncertainty. Interview material was used to verify that theorized constructs had observable behavioral counterparts in practice and to calibrate the wording of items for ecological validity. The initial pool of 40 items was constructed to cover each of the six steps, ensuring content validity by design rather than post hoc.

3.2 Iterative Refinement and Validity Checks

The item reduction and refinement proceeded through four successive stages, each focusing on a distinct validity criterion.

Stage 1. Face Validity: LLM-Assisted Pre-Test. As a cost-effective screen before expert review², we prompted OpenAI’s GPT to describe the scientific method and then asked which polar option best represented a scientific approach for each item. Items where OpenAI’s GPT did not select the theoretically designated pole were rephrased for clarity; no items were eliminated at this stage. This screen was not a validity test in itself — it served to ensure item wording was unambiguous before the expert panel reviewed content.

Stage 2. Content Validity: Expert Review and Role-Play. To assess whether the item pool covered the full SDM domain³, 16 researchers studying scientific decision-making (18 invited; 88% response rate) completed the survey under randomly assigned conditions. Seven were asked to respond as a “scientific manager”; nine were asked to respond as the “opposite of a scientific manager”. Participants then provided item-by-item feedback. We flagged items if non-response exceeded 20%, fewer than 75% chose the theoretically designated pole, the scientific-group mean fell below 4 (on a 7-point scale), the opposite-group mean exceeded 4, or the between-group mean difference fell below 2 or reversed direction. Agreement was uneven across items: on average, 59%

¹Interviews conducted February–March 2023.

²Conducted March 2023.

³Conducted April 2023.

Table 2: Item Development History

Phase	N Items	Action	Rationale
Initial pool	40	Generated from theory and interviews	Deductively grounded in the six-step SDM framework; supplemented by semi-structured interviews with 12 senior managers to ensure behavioral realism
ChatGPT pre-test	40	Polarity check; no items eliminated	For each item, ChatGPT was prompted to identify which polar option best represented scientific reasoning; polarities and wording were revised where the AI-designated pole diverged from the theoretical assignment (March 2023)
Expert review	32	Eliminated 8 items	16 researchers ($N = 18$ invited; 88% response rate) randomly assigned to respond as a “scientific” or “opposite” manager; items flagged for high non-response, wrong directionality, or insufficient group separation across 5 feedback rounds; scale reduced from 7-point to 5-point on expert recommendation (April 2023)
Pilot and nudge	28	Eliminated 4 items	Items with poor discrimination or format issues removed following internal pre-test; nudge experiment with 462 managers (354 online via Prolific, 48 elite b-school alumni, 60 firm managers) confirmed instrument sensitivity to scientific vs. intuitive priming across all subsamples (April–May 2023)
Final	24	Retained	Optimal domain coverage: 11 causal reasoning items, 9 evidence evaluation items, 4 adaptive learning items; completion time <15 min

Note. Development conducted February–May 2023. The initial item pool was derived from the six-step SDM theoretical framework and semi-structured interviews with 12 senior managers. A ChatGPT face validity check verified polarity assignments before human review. The expert panel ($N = 16$) provided item-by-item feedback across 5 rounds and recommended reducing the response scale from 7 to 5 points to lower cognitive burden. The pilot phase included an internal pre-test and a three-subsample nudge experiment ($N_{\text{total}} = 462$ managers) in which respondents were randomly assigned to answer the survey as a “scientific” manager, an “intuitive” manager, or themselves (placebo); the instrument successfully differentiated all three conditions in each subsample. The final 24-item bipolar Likert scale (1–5) was retained for full deployment.

of scientific-condition experts chose the designated pole (target: $\geq 75\%$), and the mean between-group difference was 0.95 points (target: ≥ 2.0). Only 6 of the 36 items tested met all criteria without revision. Qualitative feedback revealed three recurring problems: terminology inaccessible to managers (e.g., *assumptions* revised to *initial beliefs*; *validation* revised to *testing*), response options that were not mutually exclusive, and items that assumed access to formal documentation. These problems guided systematic rewording. Eight items were eliminated due to persistent confusion or lack of differentiation; the remaining items were revised before Stage 3. The expert panel also recommended switching from a 7-point to a 5-point scale to reduce cognitive burden.

Stage 3. Construct Validity: Nudge Experiment. To test whether the instrument captured meaningful variation under experimental manipulation⁴, 462 managers across three subsamples responded as "scientific", "intuitive", or "themselves" (placebo). 354 managers recruited via Prolific (three waves), 48 alumni of an elite European business school⁵, and 60 managers from medium-sized Italian firms were recruited. For the complete online sample, the scientific–intuitive difference was $\Delta = 0.780$ ($p < .001$), the placebo–intuitive difference was $\Delta = 0.273$ ($p < .001$) and the scientific–placebo difference was $\Delta = 0.507$ ($p < .001$). Results replicated across subsamples. Theorizing items showed larger between-group differences than empirical assessment items, consistent with theorizing being the more discretionary component of SDM.

Stage 4. Criterion Validity: Vignette Check. SDM scores correlated positively with the share of scientifically correct responses to six vignette-based behavioral questions (Pearson $r = 0.285$ – 0.614 across subsamples, all $p < .05$), providing preliminary criterion validity evidence. Formal criterion validity analysis at scale is reported in Section 4.

⁴Conducted in May 2023.

⁵Alumni were all in the placebo condition.

3.3 Final Instrument

The final 24-item SDM instrument spans the full conceptual domain. Eleven items measure causal reasoning, nine measure evidence evaluation, and four measure adaptive learning. This allocation reflects the relative cognitive complexity of each dimension: causal reasoning encompasses three interrelated processes (problem diagnosis, theory construction, and hypothesis formulation); evidence evaluation spans two (experimental design and data collection); and adaptive learning represents a single process (belief updating).

Response Format. Items use a 5-point bipolar Likert scale. Each pole represents a qualitatively distinct decision-making orientation rather than a degree of agreement. This format reduces social desirability bias by presenting both anchors as plausible managerial behaviors, placing respondents along a behavioral continuum rather than inviting endorsement of the normatively superior option. A neutral midpoint (3) is included. Higher values consistently indicate greater alignment with scientific decision-making. Respondents may select “I do not understand this question”; such responses are treated as missing.

The full text of the items appears in Appendix A1. Item-level descriptive statistics and missing data patterns are shown in Table A2 and Table A4. The inter-item correlation matrix appears in Appendix A5.

4 Validation Evidence

4.1 The Logic of Formative Construct Validation

SDM is a formative construct. Its indicators define the construct rather than reflect a single latent trait (Jarvis et al., 2003; Diamantopoulos and Winklhofer, 2001). This measurement model requires distinctive validation logic. No single statistic establishes validity. Internal consistency metrics assume interchangeable indicators and, therefore, misspecify formative constructs (Bollen

and Lennox, 1991).⁶ Global fit indices from confirmatory factor analysis impose reflective measurement assumptions that are incompatible with the data-generating process.

Validation must instead accumulate evidence across complementary criteria. We demonstrate (1) domain coverage (Section 4.2), (2) indicator non-redundancy via multicollinearity diagnostics (Section 4.3), (3) causal sensitivity to experimental manipulation (Section 4.4), and (4) structural distinguishability of subdimensions (Section 4.5).

4.2 Content Validity and Development-Stage Validation

Content validity requires that items span the theoretical domain without omission or redundancy. We operationalized SDM’s six steps and three dimensions through theory-driven definition, semi-structured interviews with 12 managers, expert review, and structured role-play protocols (Section 3). For formative measures, content validity is consequential: because indicators need not correlate strongly, systematic domain coverage at the design stage is the only safeguard against construct deficiency.

We subjected the initial item pool to three pre-data-collection checks. First, we used a large language model (GPT-4) as a low-cost discriminant pre-screen: we prompted the model to respond as a scientific, intuitive, or bureaucratic decision-maker and flagged items that produced indistinguishable response profiles across conditions for revision or removal. This step served to identify non-discriminating items efficiently before committing manager time, not to establish validity. Second, structured role-play sessions with eight managers, who enacted scientific versus intuitive decision-making scenarios while thinking aloud, led to the revision of items that failed to

⁶This logic applies within subdimensions as well as across them. A natural question is whether items within a single dimension—say, Causal Reasoning—should correlate with each other, and whether Cronbach’s alpha could therefore be applied at the subdimension level. The answer is no, for the same reason it is inappropriate at the construct level. Items within Causal Reasoning each capture a distinct cognitive act: problem diagnosis, causal theorizing, hypothesis formulation. A manager who identifies problems systematically may not formulate testable hypotheses, and a manager who formulates hypotheses may not decompose them into falsifiable predictions. These are non-interchangeable operations, not multiple proxies of a single “ability to reason causally.” The VIF diagnostics (Section 4.3) confirm this empirically: all items, including those within the same subdimension, contribute unique variance. Applying alpha within subdimensions would impose the same reflective assumption that the formative specification rejects at the construct level.

elicit distinct verbal protocols from human participants. Third, we retained only items that demonstrated sensitivity to an experimental manipulation of decision-making orientation in a separate sample (Section 4.4). These checks reduced the initial pool from 40 to 24 items. Expert review minutes and representative refinement decisions appear in Appendix A.3.

4.3 Indicator Non-Redundancy

Having established that the 24 retained items cover the theoretical domain, we ask a distinct question: Does each item contribute unique information, or do some indicators overlap to the point of redundancy? For a formative index, redundant items do not improve measurement; they inflate the weight of an arbitrarily sampled facet of the construct (Coltman et al., 2008).

We assessed indicator non-redundancy via variance inflation factors (VIF) for each item, computed by regressing each indicator on all remaining 23 items. Table 3 reports the results. VIF values range from 1.021 to 1.226, well below the conventional threshold of 5 (or even the conservative threshold of 3), with tolerance values ranging from .816 to .979, all above the .20 minimum (Hair et al., 2010). No item is collinear with the remaining set. These diagnostics confirm that each SDM indicator contributes distinct information, consistent with the formative specification and with inter-item correlations that are low by design.

4.4 Experimental Sensitivity

Non-redundant items can still fail to capture what they claim to capture. A formative instrument is credible only if its scores respond to theoretically motivated causal variation in the target construct (Shadish et al., 2002; Messick, 1995). We tested this directly by exposing 462 managers to one of three randomly assigned priming conditions: scientific reasoning, intuitive reasoning, or a placebo control. Participants completed the 24-item SDM scale after the prime.

The manipulation produced large, ordered differences across all four SDM scores (Table 4). For overall SDM, scientific priming ($M = 3.58$, $SD = 0.42$) yielded substantially higher scores than placebo ($M = 3.07$, $SD = 0.32$; $d = 1.40$), which in turn exceeded intuitive priming ($M =$

Table 3: Multicollinearity Diagnostics: Variance Inflation Factors

Item	VIF	Tol.	Status	Item	VIF	Tol.	Status
sdm23	1.19	.84	OK	sdm14	1.09	.92	OK
sdm7	1.19	.84	OK	sdm11	1.09	.92	OK
sdm9	1.16	.86	OK	sdm24	1.08	.93	OK
sdm19	1.16	.86	OK	sdm15	1.07	.93	OK
sdm20	1.15	.87	OK	sdm16	1.07	.93	OK
sdm12	1.13	.88	OK	sdm22	1.07	.94	OK
sdm13	1.11	.90	OK	sdm18	1.07	.94	OK
sdm21	1.10	.91	OK	sdm4	1.06	.94	OK
sdm2	1.10	.91	OK	sdm3	1.05	.96	OK
sdm8	1.10	.91	OK	sdm5	1.04	.96	OK
sdm17	1.10	.91	OK	sdm10	1.03	.97	OK
sdm1	1.09	.91	OK	sdm6	1.03	.97	OK

Note. Variance Inflation Factor (VIF) and tolerance (Tol.) statistics for all 24 SDM items, sorted by VIF in descending order. VIF values < 5 (or conservatively < 3) indicate acceptable multicollinearity; tolerance values $> .20$ are acceptable. All items pass conventional thresholds, suggesting minimal redundancy and appropriate item independence. $VIF = 1/(1 - R^2)$ where R^2 is from regressing each item on all other items. Tolerance = $1/VIF$.

2.80, $SD = 0.33$; placebo–intuitive $d = 0.83$). The contrast between the two active conditions was especially pronounced ($d = 2.09$). The omnibus test confirmed the overall pattern, $F(2, 459) = 175.21$, $p < .001$, $\eta^2 = .43$. The same ordering held for each subdimension: Causal Reasoning ($F = 134.11$, $p < .001$), Evidence Evaluation ($F = 84.77$, $p < .001$), and Adaptive Learning ($F = 47.89$, $p < .001$). The instrument responds to causal changes in reasoning orientation across all three dimensions, confirming that it measures reasoning processes rather than stable traits or response tendencies. Full regression specifications with robustness checks, item-level effects for all 24 indicators, replication across independent samples, attention check results, and the verbatim prime texts appear in Appendix C.

A methodological question arises from the formative necessity interpretation: if each subdimension is a non-substitutable component of SDM, should the weakest dimension—rather than the average—define the composite score? The per-dimension results address this directly. Scientific priming elevated all three subdimensions simultaneously (Causal Reasoning $d = 1.41$, Evidence

Evaluation $d = 0.78$, Adaptive Learning $d = 0.80$), ruling out the possibility that condition differences in the average mask a near-zero contribution from one component. As a further robustness check, we replace the mean with the minimum of the three subdimension scores—the most conservative aggregate under a strict necessity interpretation. Results are preserved: $F(2, 459) = 126.44$, $p < .001$, $\eta^2 = .355$, with $d = 1.11$ (scientific vs. placebo) and $d = 1.81$ (scientific vs. intuitive), versus $\eta^2 = .433$, $d = 1.40$, and $d = 2.09$ under the average. The same conclusion holds for vignette prediction and the hierarchy gradient. Appendix F reports the complete comparison across all key analyses.

Table 4: SDM Scores by Experimental Condition

Scale	Condition Means (M , SD)			Cohen's d		$F(2, 459)$
	Scientific ($n = 140$)	Placebo ($n = 183$)	Intuitive ($n = 139$)	Sci–Int	Sci–Plc	
SDM Overall	3.58 (.42)	3.07 (.32)	2.80 (.33)	2.09	1.40	175.21***
Causal Reasoning	3.54 (.51)	2.92 (.38)	2.73 (.43)	1.73	1.41	134.11***
Evidence Evaluation	3.50 (.48)	3.14 (.44)	2.76 (.50)	1.49	0.78	84.77***
Adaptive Learning	3.89 (.74)	3.32 (.67)	3.09 (.70)	1.10	0.80	47.89***

Note. *** $p < .001$. $\eta^2 = .43$ for SDM Overall. Cohen's d was computed between Scientific and Intuitive conditions (Sci–Int) and between Scientific and Placebo conditions (Sci–Plc). Wording of the priming scenarios appears in Appendix C.

4.5 Internal Structure

We estimated confirmatory factor models for two purposes: to test whether the three subdimensions are empirically distinguishable from a more parsimonious two-factor solution, and to assess whether the data are consistent with a single underlying latent factor⁷ CFA fit indices serve here as a specification test rather than a validity benchmark.

⁷Standard CFA assumes effect indicators that share a common latent cause. Applied to a formative construct, where indicators define rather than reflect the construct, CFA will produce poor fit by design; poor fit here is diagnostic, not a validity concern (Jarvis et al., 2003; Diamantopoulos and Winklhofer, 2001; Bollen and Lennox, 1991).

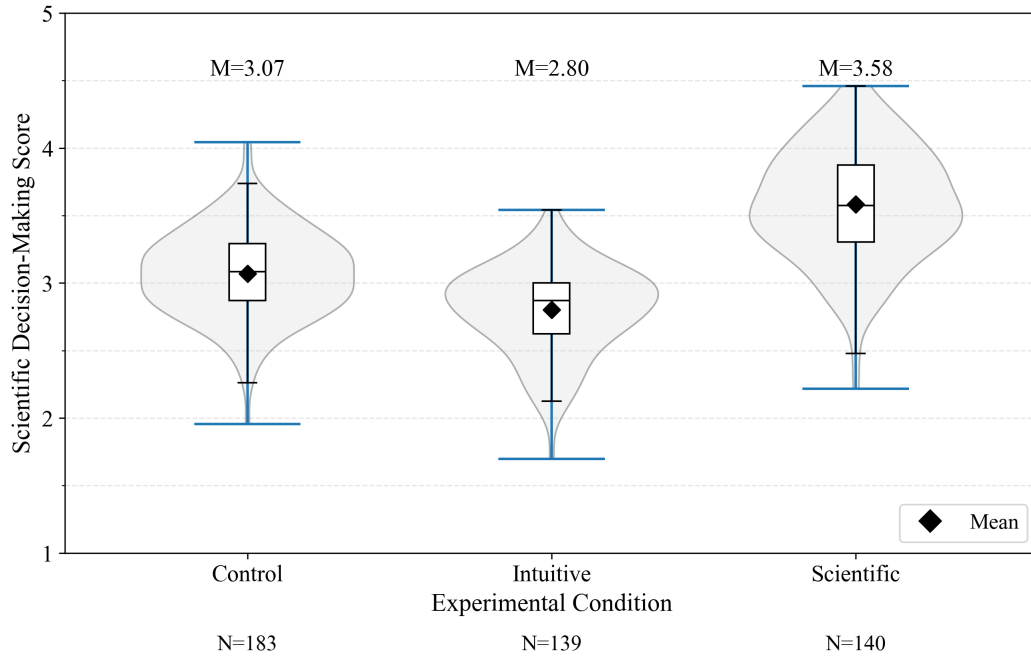


Figure 1: Priming Effects

Data consistency with a formative measurement structure SDM is specified as a formative construct: its indicators cover the theoretical domain rather than reflecting a single latent cause (Jarvis et al., 2003; Diamantopoulos and Winklhofer, 2001). Four CFA specifications were estimated (single-factor, two-factor, three-factor, and second-order hierarchical), and CFI values fall below .50 and RMSEA near .051 across all four (Appendix B2; full loadings in Appendix B1). This confirms the formative specification: SDM items capture distinct practices that a manager can perform independently. "I conduct pilot tests" and "I update beliefs when surprised" are both components of scientific decision-making, but they describe non-interchangeable acts; a manager can per one without the other. Low inter-item correlations reflect good domain coverage, not poor measurement; the VIF diagnostics in Section 4.3 ($VIF_{\max} = 1.19$, well below the threshold of 3) confirm that each indicator contributes distinct information. The three-factor solution yields negative interfactor correlations involving Causal Reasoning ($r = -.88$ with Evidence Evaluation; $r = -.48$ with Adaptive Learning). These are a modeling artifact: when CFA is applied to formative indicators, the factor covariance matrix absorbs the misfit and estimated correlations can take implausible values (Jarvis et al., 2003; Dolan, 1994). At the composite level, all three dimensions

correlate positively ($r = .12$ to $.26$); Appendix B5 reports the contrast alongside robustness checks on item coding.

Three factors v.s. two factors Each additional factor yields a significant improvement in fit ($\Delta\chi^2 = 48.35$, $\Delta df = 1$, $p < .001$ for the two-factor vs. one-factor comparison; $\Delta\chi^2 = 54.74$, $\Delta df = 2$, $p < .001$ for the three-factor vs. two-factor comparison), indicating that each theorized dimension captures non-redundant variance. The distinction between Evidence Evaluation and Adaptive Learning deserves particular attention given their high estimated interfactor correlation ($r = .95$). The distinction is substantive: a manager can evaluate evidence rigorously without updating prior beliefs in light of it, as motivated reasoning and commitment escalation each illustrate (Nickerson, 1998; Bazerman and Moore, 2013). The deployment data provide behavioral confirmation: Adaptive Learning differentiates organizational tiers meaningfully whereas Evidence Evaluation does not (AL gradient, CEO vs. frontline = $+.168$ scale points; EE gradient = $-.013$; difference = $.181$, $p < .01$). Two empirically equivalent constructs would produce statistically indistinguishable tier gradients.

The Adaptive Learning subscale contains four items. For a formative construct, item count is not the relevant criterion; domain coverage without redundancy is (Diamantopoulos and Winklhofer, 2001). Belief adjustment involves four non-interchangeable cognitive operations: recognizing when incoming evidence is diagnostically relevant, revising prior probabilities in proportion to that evidence, acknowledging surprises and novel states, and revising beliefs in subsequent decision steps (Karni and Schmeidler, 2022). The four items map onto these operations without overlap; no pair describes the same cognitive act. Adding items would require either repeating these operations, introducing redundancy, which is inconsistent with formative specification, or expanding the domain beyond belief updating into content already captured by Evidence Evaluation or Causal Reasoning. The four-item count reflects deliberate domain coverage under the formative logic, not a shortfall relative to conventional scale development norms.

5 Application

We apply the SDM scale to a sample of 9,458 managers across 61 firms to examine how scientific decision-making is distributed within organizations and whether it is associated with vignette performance. SDM is individually distributed within firms, follows the organizational hierarchy, and is associated with both individual and collective vignette performance. Descriptive statistics for all variables appear in Table E.1 (Appendix E).

5.1 Level of Analysis: SDM as an Individual-Level Construct

Before examining how SDM varies across organizational levels, we establish the level at which SDM operates. Random intercept models estimated via REML decompose the total SDM variance into between-firm and within-firm components for each subdimension (Table 5). Between-firm variance accounts for 1.1% of total variance in SDM overall, with subdimension ICCs ranging from 0.7% (Causal Reasoning, Adaptive Learning) to 2.0% (Evidence Evaluation). All values fall well below the conventional 5% threshold for meaningful group-level clustering. The pattern matches the construct definition. SDM captures individual cognitive dispositions (habitual tendencies to seek disconfirming evidence, update priors, and construct causal models) that form through education, deliberate practice, and epistemic socialization rather than through firm membership (Stanovich, 2009; Camuffo et al., 2020b). Firms do not appear to impose a sufficiently uniform reasoning environment to make managers within a firm more similar to each other than to managers in other firms. The within-firm variance that does exist is not random: it follows the organizational hierarchy.

5.2 The Hierarchical Architecture of SDM

Scientific decision-making is not uniformly distributed within firms: it follows the organizational hierarchy. Table 6 reports OLS regressions of SDM scores on the hierarchical tier, with Tier 5 (Frontline) as the reference category. C-suite and top management score significantly higher than

Table 5: Between-Firm vs. Within-Firm Variance in SDM (ICC)

Construct	Variance		ICC	Between (%)	N	Firms
	Between-firm	Within-firm				
SDM Overall	0.0009	0.0843	0.011	1.1%	9,458	61
Causal Reasoning	0.0009	0.1379	0.007	0.7%	9,452	61
Evidence Evaluation	0.0036	0.1786	0.020	2.0%	9,456	61
Adaptive Learning	0.0028	0.3702	0.007	0.7%	9,449	61

Note. Random intercept model ($Y_{ij} \sim 1 + (1|\text{firm})$) estimated via REML. $\text{ICC}(1) = \sigma_{\text{between}}^2 / (\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2)$. Between (%) is the proportion of total variance attributable to firm membership. All ICCs fall below the conventional threshold of 5%, indicating that SDM operates primarily as an individual-level construct.

frontline managers on overall SDM ($b = .07, p < .001$), and the gradient is monotonic: each tier below scores progressively lower. This pattern survives firm fixed effects, ruling out the possibility that it merely reflects differences in firm composition.

The gradient is not uniform across the subdimensions (Figure 2). Adaptive Learning drives the strongest hierarchical differentiation: top management scores exceed frontline scores by .19 percentage points ($p < .001$), and the gap narrows steadily at each tier. Causal Reasoning shows a similarly clean gradient. Evidence Evaluation, by contrast, does not differentiate tiers; coefficients are small and not significantly different from zero at all levels, suggesting that the capacity to evaluate evidence is not systematically selected or developed across the hierarchy. The within-firm SDM pattern thus reflects hierarchy in *how* managers reason, not in *whether* they seek evidence.

This dissociation between Adaptive Learning and Evidence Evaluation is not merely descriptive. It provides behavioral evidence that the two subdimensions are empirically distinct constructs. The Adaptive Learning gradient ($\Delta = .19$ from Tier 1 to Tier 5) exceeds the Evidence Evaluation gradient ($\Delta = .02$) by .17 scale points ($SE = .04, p < .001$). If the two dimensions captured the same underlying disposition, as a high CFA interfactor correlation of $r = .95$ might suggest, their tier gradients would be statistically indistinguishable.

Table 6: Hierarchical Level Effects

	Overall		Causal		Evidence		Adaptive	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A: Tiers Only</i>								
T1 (C-suite)	0.05*** (0.01)	0.07*** (0.02)	0.04* (0.02)	0.07** (0.02)	0.01 (0.02)	0.02 (0.02)	0.14*** (0.03)	0.19*** (0.03)
T2 (Senior)	0.01 (0.01)	0.05** (0.02)	0.03* (0.01)	0.08*** (0.02)	-0.05** (0.02)	-0.01 (0.02)	0.05* (0.02)	0.07* (0.03)
T3 (Middle)	0.02** (0.01)	0.04*** (0.01)	0.03** (0.01)	0.05*** (0.01)	-0.02 (0.01)	-0.00 (0.02)	0.06*** (0.02)	0.08*** (0.02)
T4 (Junior)	0.01 (0.01)	0.02* (0.01)	0.01 (0.01)	0.02* (0.01)	-0.01 (0.01)	0.00 (0.01)	0.06*** (0.02)	0.07*** (0.02)
<i>Panel B: + Education</i>								
T1 (C-suite)	0.05*** (0.01)	0.06*** (0.02)	0.04* (0.02)	0.07** (0.02)	0.00 (0.02)	0.01 (0.02)	0.13*** (0.03)	0.18*** (0.03)
T2 (Senior)	0.01 (0.01)	0.04** (0.02)	0.03 (0.01)	0.07*** (0.02)	-0.06*** (0.02)	-0.02 (0.02)	0.04 (0.02)	0.07* (0.03)
T3 (Middle)	0.02* (0.01)	0.03*** (0.01)	0.02* (0.01)	0.05*** (0.01)	-0.02 (0.01)	-0.00 (0.02)	0.06*** (0.02)	0.08*** (0.02)
T4 (Junior)	0.01 (0.01)	0.02* (0.01)	0.01 (0.01)	0.02* (0.01)	-0.01 (0.01)	0.00 (0.01)	0.05** (0.02)	0.07*** (0.02)
<i>Panel C: + Tenure</i>								
T1 (C-suite)	0.05*** (0.01)	0.06*** (0.02)	0.04* (0.02)	0.07** (0.02)	0.00 (0.02)	0.01 (0.02)	0.14*** (0.03)	0.18*** (0.03)
T2 (Senior)	0.01 (0.01)	0.04** (0.02)	0.03* (0.01)	0.07*** (0.02)	-0.06*** (0.02)	-0.02 (0.03)	0.04 (0.02)	0.07* (0.03)
T3 (Middle)	0.02* (0.01)	0.04*** (0.01)	0.03** (0.01)	0.05*** (0.01)	-0.02 (0.01)	-0.00 (0.02)	0.06*** (0.02)	0.08*** (0.02)
T4 (Junior)	0.01 (0.01)	0.02* (0.01)	0.01 (0.01)	0.02* (0.01)	-0.01 (0.01)	0.00 (0.01)	0.05** (0.02)	0.07*** (0.01)
<i>Panel D: + Decision</i>								
T1 (C-suite)	0.05*** (0.01)	0.07*** (0.02)	0.04* (0.02)	0.07** (0.02)	0.02 (0.02)	0.02 (0.02)	0.15*** (0.03)	0.19*** (0.03)
T2 (Senior)	0.02 (0.01)	0.05** (0.02)	0.03* (0.02)	0.08*** (0.02)	-0.05** (0.02)	-0.01 (0.03)	0.06* (0.02)	0.07* (0.03)
T3 (Middle)	0.03** (0.01)	0.04*** (0.01)	0.03** (0.01)	0.05*** (0.01)	-0.01 (0.01)	0.01 (0.02)	0.07*** (0.02)	0.08*** (0.02)
T4 (Junior)	0.01 (0.01)	0.02** (0.01)	0.01 (0.01)	0.02* (0.01)	-0.00 (0.01)	0.01 (0.01)	0.06*** (0.02)	0.07*** (0.02)
Firm FE	No	Yes	No	Yes	No	Yes	No	Yes
<i>N</i>	9,458		9,452		9,456		9,449	
Firms	61		61		61		61	

Note. Dependent variables: SDM Overall, Causal Reasoning, Evidence Evaluation, Adaptive Learning. Odd columns: OLS with robust SEs. Even columns: OLS with firm fixed effects and SEs clustered at the firm level. Standard errors in parentheses. Reference category: Tier 5 (Frontline). Panel B adds education controls (STEM background, degree level). Panel C adds tenure. Panel D adds decision type fixed effects. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

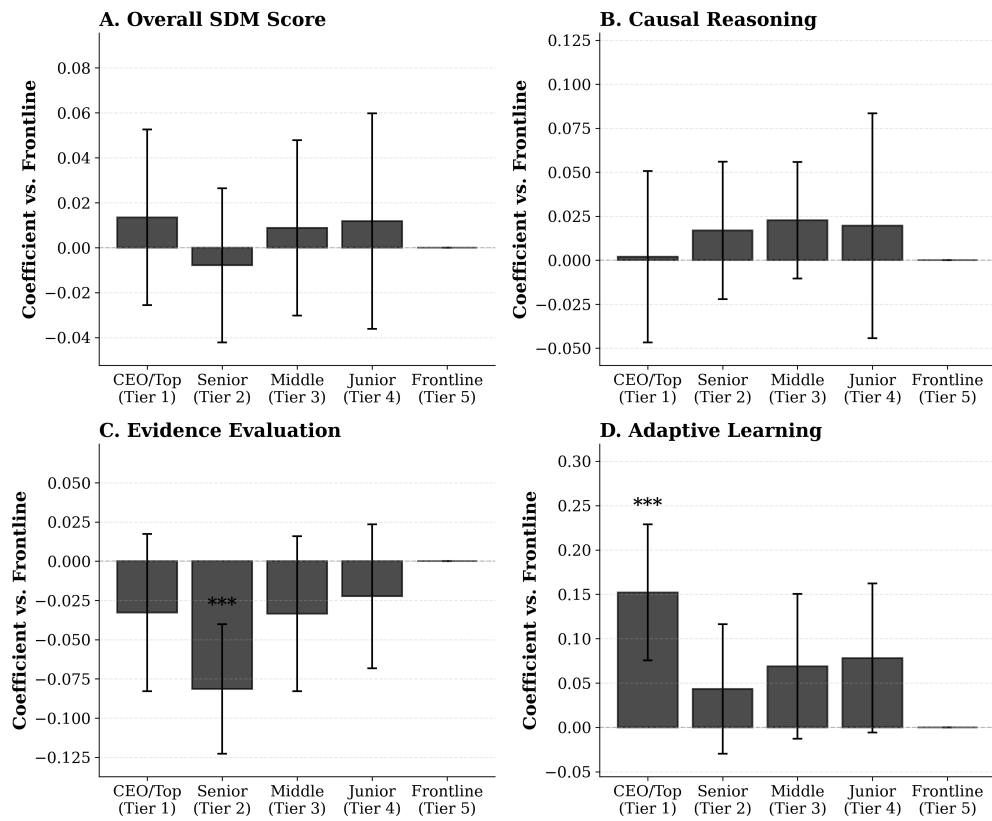


Figure 2: OLS Coefficient Scores by Hierarchical Tier and Subdimension

5.3 SDM and Individual Vignette Performance

The vignette task captures how consistently managers apply the scientific reasoning method to realistic decision scenarios: responses are scored for alignment with systematic evidence-seeking, causal theorizing, and belief updating, not for their ability to predict favorable outcomes. Methodologically consistent reasoning and good outcomes are related but distinct; the vignette measures the former.

SDM scores are associated with performance on this vignette task. Table 7 reports progressive OLS models with vignette accuracy as the outcome. In the baseline model, a one-unit increase in the overall SDM score is associated with a .12 increase in the proportion of responses consistent with scientific decision-making ($p < .001$). The coefficient is stable across specifications that add hierarchical tier, education, tenure, and decision area controls (range: .117-.122), indicating that the SDM-performance relationship is not confounded by seniority or human capital. Pearson correlations with the three subdimensions are .12, .11, and .15 for Causal Thinking, Evidence-Based Thinking, and Adaptive Experimentation, respectively.

Figure 3 illustrates this relationship by SDM quintile. Managers in the upper quintile ($\text{SDM} \geq 3.47$) give responses consistent with scientific decision-making on 54.7% of vignette scenarios, compared to 44.3% for managers in the lower quintile ($\text{SDM} \leq 2.96$) – a difference of approximately 12 percentage points.

5.4 SDM and Collective Vignette Performance

The SDM–performance relationship extends to the firm level. Table 8 reports firm-level OLS regressions ($N = 57$ firms with ≥ 5 respondents) predicting mean vignette accuracy from firm mean SDM. Firms whose managers score higher on SDM on average achieve higher collective decision accuracy ($b = .277$, $p = .013$, $R^2 = .107$). Within-firm SDM dispersion (the standard deviation of individual SDM scores) does not add predictive power beyond the mean ($b = .120$, $p = .36$; M2). The result indicates that, at the organizational level, what matters is the *level* of scientific reasoning, not its homogeneity across members.

Table 7: Relationship Between SDM Scores and Vignette Performance

	Dependent Variable: Vignette Score (Share Correct)							
	Overall SDM		Causal Reasoning		Evidence Evaluation		Adaptive Learning	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pearson correlation	0.175***		0.120***		0.111***		0.150***	
SDM score	0.122*** (0.007)	0.120*** (0.010)	0.066*** (0.006)	0.064*** (0.007)	0.053*** (0.005)	0.053*** (0.005)	0.050*** (0.003)	0.048*** (0.003)
+ Hierarchical level	0.121*** (0.007)	0.118*** (0.009)	0.064*** (0.006)	0.062*** (0.007)	0.054*** (0.005)	0.053*** (0.005)	0.049*** (0.003)	0.047*** (0.003)
+ Education	0.120*** (0.007)	0.117*** (0.009)	0.063*** (0.006)	0.061*** (0.006)	0.054*** (0.005)	0.052*** (0.005)	0.048*** (0.003)	0.047*** (0.003)
+ Tenure	0.120*** (0.007)	0.117*** (0.009)	0.063*** (0.006)	0.061*** (0.006)	0.054*** (0.005)	0.052*** (0.005)	0.048*** (0.003)	0.047*** (0.003)
+ Decision area	0.119*** (0.007)	0.117*** (0.008)	0.062*** (0.006)	0.061*** (0.006)	0.054*** (0.005)	0.053*** (0.005)	0.047*** (0.003)	0.046*** (0.003)
R^2	0.044		0.028		0.028		0.035	
Within R^2	0.059		0.044		0.044		0.051	
Firm Fixed Effects	No	Yes	No	Yes	No	Yes	No	Yes
Observations	9,458		9,452		9,456		9,449	
Firms	61							

Note. Progressive model specifications testing the relationship between SDM dimensions and vignette performance (proportion of correct responses in scenario-based decision tasks). Pearson correlation shown in first row. Each SDM dimension is shown with OLS (odd columns) and firm fixed effects (even columns). Models progressively add: (1) SDM score only; (2) hierarchical level controls (5-tier classification from CEO/Top to Frontline); (3) education controls (STEM background, graduate degree); (4) tenure (years worked at the firm); (5) decision area fixed effects (corporate strategy, competitive strategy, internal transformation, external stakeholders, human resources). Robust standard errors in parentheses. Standard errors clustered at firm level for firm FE models. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

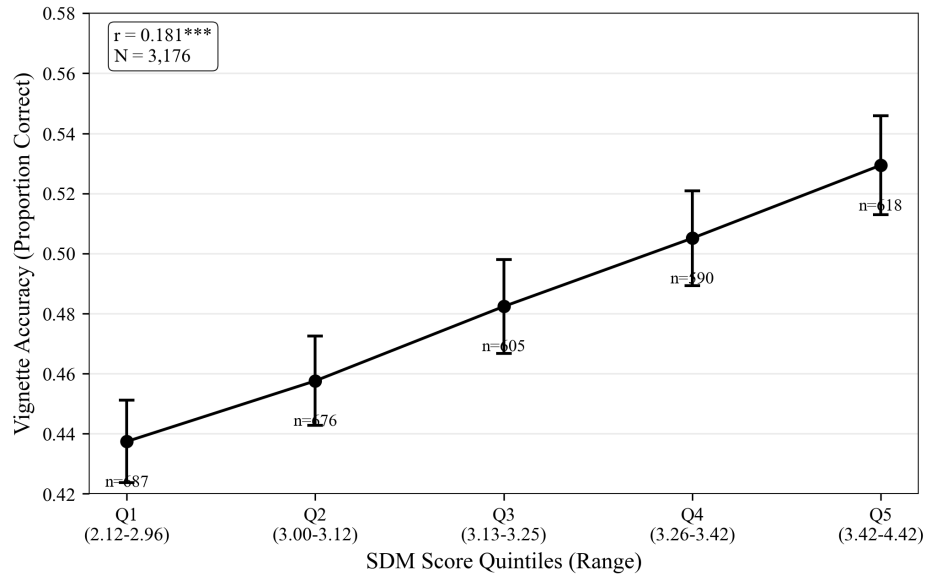


Figure 3: SDM by Quintile and Vignette Accuracy

Table 8: Firm-Level SDM and Collective Decision Quality

	M1 (1)	M2 (2)	M3 (3)
Firm mean SDM	0.277* (0.107)	0.280* (0.107)	0.281* (0.109)
Firm SD of SDM		0.120 (0.127)	0.119 (0.128)
Firm <i>N</i> (managers)			0.000 (0.001)
R^2	0.107	0.121	0.122
Firms		57	

Note. OLS regressions at the firm level. Dependent variable is firm mean vignette accuracy (proportion correct). Sample restricted to firms with ≥ 5 respondents. Firm mean SDM is the within-firm average of individual SDM scores (scale: 1–5). Firm SD of SDM is the within-firm standard deviation of individual SDM scores. Standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

6 Limitations

Six limitations qualify the evidence above.

Criterion validity: modest effect and same-team design. SDM correlates with vignette accuracy at $r = .175$ ($p < .001$), explaining approximately 3% of variance. The modest magnitude is expected: vignette performance reflects contextual knowledge, working memory, and risk preferences in addition to reasoning orientation, so $r = .175$ understates the SDM-specific signal. We treat the vignette as a partial index of methodological reasoning orientation, not a measure of decision outcomes or firm performance.

A more substantive concern is that the same research team developed both the SDM scale and the vignettes. Circularity is not a problem if the two instruments were developed independently of each other, which they were: the vignettes were constructed from real strategic scenarios with responses evaluated for alignment with scientific reasoning principles, with no reference to SDM item content. But an external reader cannot verify that separation from the published record alone. The appropriate remedy is validation against an independently developed criterion; that evidence is not yet available. Until it is, the priming experiment provides the primary validity evidence: it used a separate sample, a manipulation with no connection to scale development, and a dependent measure designed entirely independently ($d = 2.09$, $\eta^2 = .43$).

A related design feature concerns administration order. The SDM scale always preceded the vignette; the order was never counterbalanced. Completing 24 items describing scientific reasoning practices may prime respondents to approach the vignette more deliberately. If this priming is stronger for low-SDM managers—who would otherwise rely more on intuition—it artificially narrows the performance gap between high- and low-SDM respondents and attenuates the observed correlation. Under this interpretation, $r = .175$ is a conservative lower bound: counterbalancing would restore natural between-person variation in vignette performance and likely reveal a stronger association.

Cross-sectional design. The deployment study is cross-sectional. We cannot establish whether SDM predicts decision quality over time or whether the hierarchical gradient reflects selection, socialization, or role-specific experience. Longitudinal data linking instrument scores to decision outcomes over multiple periods would address both questions.

Sample self-selection. Participating organizations elected to join the SDM Project and agreed to survey their managers. Firms that opt into such initiatives may be more analytically oriented than the general organizational population, potentially compressing the lower tail of the SDM distribution and attenuating between-firm variance. The ICC evidence (less than 2% between-firm variance) is consistent with this possibility, though it is also consistent with SDM being a genuinely individual-level attribute. Replication in a probability sample of organizations would resolve this ambiguity.

Firm-specific tenure versus career experience. The tenure measure in this study captures years with the current firm, not total years of professional experience. These are distinct constructs: firm-specific tenure indexes exposure to a particular organizational environment, whereas career experience indexes cumulative task practice across contexts. The weak negative correlation between tenure and SDM ($r = -.044$) is theoretically interpretable under cognitive entrenchment accounts (Dane, 2010), but it conflates both mechanisms. Future work should separate firm-specific and career-level tenure to test whether the negative pattern reflects organizational lock-in, declining openness to new methods with age, or both.

Scope of performance claims. This paper establishes that SDM is associated with higher vignette performance—responses more consistent with the scientific approach to decision-making—not with business outcomes or economic returns. The vignette measures procedural alignment with scientific reasoning; it does not capture whether decisions produce superior organizational results. The pathway from individual reasoning orientation to firm performance runs through implementation fidelity, team dynamics, and market context, none of which this study models. Whether SDM

translates into field decision outcomes or firm-level returns is an open empirical question that this paper does not claim to answer.

7 Implications

The SDM instrument transforms scientific decision-making from an experimentally induced treatment into an observable characteristic of managers. Previous work established that SDM training improves decision quality (Camuffo et al., 2020a, 2024b); this paper enables the next step, studying who already practices SDM, how it is distributed, and what its distribution reveals about how organizations structure reasoning.

7.1 Theoretical Implications

The instrument shifts the central research question from whether SDM improves outcomes to when, why, and for whom it does so. SDM is now measurable as an individual difference, opening up a systematic inquiry into its antecedents and consequences. The available correlates, education, STEM background, and hierarchical position, are consistent with theoretical expectations but do not establish causal priority. Longitudinal and experimental designs can now test whether formal training increases SDM scores durably and whether these gains translate into field decision quality.

The dimensional structure adds a second theoretical implication. The hierarchical gradient is not uniform: Adaptive Learning differentiates managerial tiers more sharply than Evidence Evaluation. One interpretation is that Adaptive Learning is more sensitive to accumulated decision experience, as senior managers have faced repeated cycles of evidence, expectation violation, and belief revision, while Evidence Evaluation may reflect a more stable cognitive style less responsive to positional experience. This asymmetry implies that hierarchy structures *which* components of scientific reasoning develop, not merely the overall level. Future research can test this interpretation by comparing SDM profiles across tenure levels and decision-domain exposure independent of formal rank.

7.2 Methodological Implications

The validation strategy offers a template for researchers developing process-based constructs with formative structure. Classical validation logic (factor analysis, internal consistency, convergent validity matrices) assumes that indicators reflect a common latent cause. For constructs whose components define rather than reflect the construct, this template is logically inappropriate (Jarvis et al., 2003; Diamantopoulos and Winklhofer, 2001). The alternative demonstrated here uses five complementary criteria: domain coverage via expert review, non-redundancy via VIF diagnostics, causal sensitivity via experiment, structural distinguishability via CFA specification tests, and behavioral prediction via an external criterion. Each criterion addresses a distinct validity threat; collectively, they accumulate evidence that no single test can provide.

7.3 Empirical Applications

The instrument enables three classes of empirical work that were previously infeasible. First, large-sample observational research: deployment across 9,458 managers in 61 firms demonstrates field-scale feasibility. SDM can now be studied as a predictor of decision outcomes, a moderator of strategy-performance relationships, or an antecedent of firm-level learning, none of which were tractable at the observational scale with prior experimental designs. The ICC between firms of 1.1% confirms that there is organizational variation to study, even if individual-level variance dominates.

Second, intervention and training research: because SDM responds to experimental priming, the instrument serves as both a pre/post diagnostic and a mechanism variable in training studies. Researchers can now ask whether SDM-improving interventions operate through specific subdimensions (Causal Reasoning vs. Adaptive Learning), allowing for more targeted program design.

Third, longitudinal designs: the instrument enables tracking SDM trajectories across career stages, organizational transitions, and firm-level experiments. Cross-sectional evidence shows that senior managers score higher; whether this reflects selection into positions, developmental

trajectories, or both is an open question that panel data can now address.

7.4 Research Agenda

Several questions become tractable. On antecedents: do training programs durably raise SDM scores, and are effects heterogeneous by role, sector, or prior reasoning style? On contingencies: when is SDM most predictive of decision quality: under high versus low uncertainty, in dynamic versus stable industries, or in decision domains with well-defined versus ambiguous feedback? On consequences: does individual and team-level SDM predict decision outcomes in naturalistic settings? Firm-level effects are a longer-term question given the limited between-firm ICC; individual and team-level outcomes are more tractable first tests. On dynamics: do SDM scores change predictably across career stages, and which dimensions are most responsive to deliberate organizational investment?

A priority for instrument development is convergent validity against established cognitive style measures. The deployment study used contextual proxies available at organizational scale; direct validation against the Need for Cognition (Cacioppo and Petty, 1982), Actively Open-Minded Thinking (Stanovich and Toplak, 2016), and epistemic curiosity (Litman et al., 2005) in a dedicated subsample would allow a multimethod-multitrait analysis and position SDM within the broader landscape of individual differences in reasoning.

The instrument does not resolve these questions, and that is not its purpose. It creates the measurement infrastructure that makes them answerable.

8 Conclusion

Scientific decision-making can be measured at scale in organizations. A 24-item instrument, validated across five complementary criteria, operationalizes SDM as a formative construct with three dimensions (Causal Reasoning, Evidence Evaluation, and Adaptive Learning) and applies across 9,458 managers in 61 organizations. Deployment reveals that SDM is not uniformly distributed: it

follows the organizational hierarchy, differentiates hierarchical tiers non-uniformly across dimensions, and predicts vignette decision accuracy.

The empirical patterns we presented carry theoretical content beyond validation. The hierarchical gradient (CEOs and senior managers scoring higher than frontline workers, net of firm composition) indicates that organizational position is associated with how managers structure reasoning under uncertainty. The dimensional asymmetry sharpens this: Adaptive Learning differentiates tiers more sharply than Evidence Evaluation, implying that positional experience develops specific components of scientific reasoning rather than elevating all dimensions uniformly. Producing these patterns required field-scale measurement that experimental designs alone cannot provide; the instrument creates that capability.

By making scientific reasoning observable and comparable at scale, the instrument enables questions that prior experimental designs could not address: who develops SDM, under what conditions it predicts outcomes, and how organizations shape the scientific approach of their managers. This paper does not resolve them. It makes them researchable.

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A Survey Instrument

A.1 Item specification

Item	Step / Question	Stem	1 (Left Anchor)	5 (Right Anchor)
Panel A. Scientific Decision-Making (SDM) Survey Items				
sdm1	Problem Diagnosis	I mostly focus on defining...	Solutions to a problem	Problems to solve
sdm2	Problem Diagnosis	I tend to reflect about why...	Problems are important	Actions are important
sdm3	Causal Theorizing	I initially focus on thinking about...	A main plan	Alternative plans
sdm4	Evidence Gathering	I am prone to prepare by gathering...	Resources	Information
sdm5	Causal Theorizing	I am mostly worried about...	Not having taken enough action	Not having reflected enough
sdm6	Problem Diagnosis	The first thing I worry about is...	The payoffs of a decision	If the decision deserves my attention
sdm7	Evidence Appraisal	Before creating a plan, I mostly examine information...	Without pre-defined procedures	Following pre-defined procedures
sdm8	Causal Theorizing	I am prone to...	Plan without discussing my assumptions	Make my assumptions explicit before planning
sdm9	Causal Theorizing	As I first think about a plan, I consider...	A single scenario of the future	Multiple scenarios of the future
sdm10	Hypothesis Testing	I mainly think about verifying plans by...	Breaking down the problem and solutions	Considering the problem and solutions as a whole
sdm11	Hypothesis Testing	I am inclined to validate...	The broad aspects of plans	Precise aspects of plans
sdm12	Hypothesis Testing	I tend to assess if...	The general description of a plan makes sense	Cause-and-effect relationships within a plan make sense
sdm13	Hypothesis Testing	As I finalize my planning, I am trying to...	Focus on fewer scenarios to consider	Search for even more scenarios to consider
sdm14	Evidence Gathering	When there is uncertainty, I mainly evaluate a plan by...	Leveraging existing information	Collecting new information
sdm15	Evidence Gathering	When thinking about a plan, I am prone to assume that...	I have enough information	I do not have enough information
sdm16	Evidence Appraisal	I establish the criteria to make a decision...	Before collecting information	After collecting information
sdm17	Evidence Appraisal	I tend to assess the outcomes of a plan...	In absolute terms (e.g., focusing on its own payoff)	In relative terms (e.g., comparing payoff with a benchmark)
sdm18	Evidence Appraisal	Typically, my initial perspective when considering a plan is...	Not neutral with respect to the plan (either favorable or unfavorable)	Neutral with respect to the plan

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Item	Step	Question	1 (Left Anchor)	5 (Right Anchor)
sdm19	Evidence Appraisal	To a large extent, the reason I collect information is to...	Confirm if a potential plan is adequate	Challenge the adequacy of a potential plan
sdm20	Evidence Appraisal	When I collect new information to assess a plan, I mainly interpret it...	Without pre-defined criteria	Based on pre-defined criteria
sdm21	Belief Adjustment	When new information contradicts my initial thoughts and plan, I...	Attempt to validate the plan again	Reconsider the plan
sdm22	Belief Adjustment	When new information contradicts my initial thoughts and plan, continuing with the original plan is...	Easy	Difficult
sdm23	Belief Adjustment	When evaluating a plan, I am inclined to find reasons to...	Support it	Challenge it
sdm24	Belief Adjustment	Mostly, final decisions are aligned with...	My initial intuition	A revised line of thought

Panel B. Decision-Making Vignette

Scenario: Imagine you are the CEO of a medium-sized retail company. You are facing pressure from the board. The company has experienced successive declines in year-to-year sales for four consecutive quarters. As the CEO, what would you do?

vignette1 –	What would you initially focus on during the next meeting with the top management team?	Brainstorm potential actions to increase sales.	Investigate potential reasons behind the decline in sales.
vignette2 –	After meeting with the top management team, what would you prioritize?	Creating a plan to revamp sales swiftly, focusing on how to best use available resources.	Creating alternative plans addressing different potential drivers of sales decline.
vignette3 –	What would you assess before engaging in any action?	The resources that could become available to the company in the future.	The adequacy of assumptions underlying one or multiple plans.
vignette4 –	What information would you collect to inform your decision?	In-house information about the stores (e.g., sales, personnel).	Tests of whether specific planned actions could work (e.g., pilot-tests).
vignette5 –	How would you interpret the information you collected?	Flexibly as it arrived.	Applying predefined criteria to assess plans.
vignette6 –	What would you do if you received evidence contradicting your initial thoughts and plan?	Continue collecting information until sufficient evidence supports the initial plan.	Reconsider the plan to suit the new evidence.

Note. Panel A reports the 24-item Scientific Decision-Making (SDM) instrument. Items are measured on a five-point bipolar scale anchored by the statements shown in the fourth and fifth columns (1 = left anchor; 5 = right anchor). Higher values consistently indicate greater alignment with scientific decision-making. The SDM construct is

specified as formative: items represent distinct but complementary steps of the scientific reasoning process (problem diagnosis, causal theorizing, hypothesis testing, evidence gathering, evidence appraisal, and belief adjustment) and are therefore not expected to exhibit high internal consistency. Panel B reports the six-item vignette-based behavioral assessment. Each question presents three response options; the scientifically grounded response is coded as correct (=1) and alternative responses as incorrect (=0). The vignette score equals the proportion of correct responses across the six items.

A.2 Item-Level Descriptive Statistics

Table A2: Item-Level Descriptive Statistics

Item	Mean	SD	Min	Max	Missing (%)	<i>N</i>
sdm1	2.39	1.22	1	5	0.86	9,387
sdm2	2.52	1.16	1	5	2.09	9,270
sdm3	2.58	1.23	1	5	0.62	9,409
sdm4	4.19	0.85	1	5	0.92	9,381
sdm5	3.10	1.16	1	5	3.64	9,123
sdm6	2.71	1.18	1	5	2.19	9,261
sdm7	3.12	1.27	1	5	0.56	9,415
sdm8	3.88	0.96	1	5	1.25	9,350
sdm9	4.19	0.79	1	5	0.49	9,422
sdm10	3.25	1.27	1	5	1.28	9,347
sdm11	3.00	1.21	1	5	1.35	9,340
sdm12	3.52	1.12	1	5	1.90	9,288
sdm13	2.87	1.24	1	5	0.84	9,388
sdm14	3.64	1.05	1	5	0.40	9,430
sdm15	3.25	1.11	1	5	0.81	9,391
sdm16	2.29	1.12	1	5	0.49	9,422
sdm17	3.44	1.11	1	5	2.36	9,245
sdm18	3.33	1.17	1	5	3.65	9,122
sdm19	2.92	1.24	1	5	1.07	9,367
sdm20	2.88	1.18	1	5	1.10	9,364
sdm21	3.89	0.97	1	5	0.46	9,424
sdm22	3.85	0.94	1	5	1.45	9,331
sdm23	3.12	1.20	1	5	0.56	9,415
sdm24	3.12	1.16	1	5	0.84	9,388
Overall	3.18	1.10	1	5	1.38	9,308

Note. Descriptive statistics for all 24 items in the Scientific Decision-Making (SDM) scale. Items measured on a 5-point bipolar Likert scale (1 = least scientific, 5 = most scientific) with an explicit neutral midpoint (3) and "Don't know/Not applicable" option. Missing percentages are calculated relative to total sample size ($N = 9,467$). Overall statistics represent the mean across all items.

A.3 Expert Review

We structured the review in two phases. Phase 1 (January–March 2023) collected iterative expert feedback on item formulation and construct validity. Phase 2 (April 11, 2023) deployed a pilot survey ($n = 16$ expert respondents) to assess question clarity, cognitive burden, response option exclusivity, and measurement biases. Below we report representative feedback and our refinement decisions.

A.3.1 Phase 1: Expert Item Review

Early-Stage Motivation (Problem Definition Block)

Original Item: “Before starting the focal project/initiative, the key stakeholders had discussed and mapped the fundamental challenge/opportunity that motivated an action by the firm.”

Expert Feedback: Reviewers identified three ambiguities. First, “key stakeholders” lacked specificity—respondents could not identify which actors were relevant. Second, the item did not clarify whether discussions included the respondent. Third, “an action” was too generic to anchor responses to specific organizational activities.

Refinement: We revised the item to specify the respondent’s involvement (“the project team including yourself”), concrete organizational terminology (“strategic initiative or operational change”), and clearer temporal anchoring. This improved respondent anchoring while maintaining construct breadth.

Original Item 3: “... alternative paths to address the same fundamental challenge/opportunity.”

Expert Feedback: One reviewer recommended replacing abstract “paths” with organizational terminology: “I would call it ‘projects/initiatives’.”

Refinement: We adopted the suggestion, revising to “alternative projects/initiatives.”

Reasoning About the Project (Theory Development Block)

Original Items 7–9: Items asked respondents about “scenarios” defined as combinations of relevant dimensions, likelihood assessments, and ideal conditions for success.

Expert Feedback: One reviewer expressed fundamental confusion: “I would not know how to answer this point. To me it would be unclear what a scenario is and how to define the likelihood of resources/conditions/activities relevant for...” This revealed a critical construct validity issue.

Refinement: We eliminated the abstract term “scenario” and restructured items to ask directly about: (1) identification of key success factors, (2) assessment of controllable factors, and (3) evaluation of required conditions. We added explanatory preambles defining how combinations of resources, conditions, and activities create distinct pathways to project success.

Expert Feedback on Item Differentiation: One reviewer noted insufficient linguistic differentiation: “The alternatives here seem small variations on the same theme. You may want to highlight more the different content of the question that you are asking by changing the wording more and highlight the differences in the text.”

Refinement: We increased differentiation between items 10–15 by using distinct action verbs (e.g., “identified,” “justified,” “prioritized”), restructuring item stems to emphasize different aspects (dimensions vs. relationships vs. scenarios), and adding parenthetical examples clarifying each item’s unique focus.

Plan to Validate Reasoning (Experiment Design Block)

Original Section Header: “Plan to validate/challenge the baseline reasoning (designing the ‘experiment’)”

Expert Feedback: One reviewer noted: “I understand that this refers to the previous question. However, in the title of the question, you may think of being more precise about what this ‘baseline reasoning’ is, otherwise the respondent has to go back to the previous question to understand what that was about.”

Refinement: We revised the header to: “Plan to test initial assumptions about success factors (designing the validation approach).” This eliminated reference to “baseline reasoning” requiring retrospective lookup, explicitly connected to “success factors” from previous sections, and replaced “experiment” with “validation approach” to match managerial vocabulary.

Analyzing Evidence (Experiment Results Block)

Original Items 21–27: “If an external person had access to the documents reporting the data collection and data analysis used to validate the baseline reasoning of the key stakeholders, she (or he) would be able to recover information on...”

Expert Feedback: One reviewer identified a critical measurement assumption: “do you want to know whether they have the documents or whether, irrespective of having any documentation, firms do any of the items below? Here it seems that you are assuming that there is a documentation about what they do, which I would not give for granted. The risk is that if there is no documentation, the respondent may report none of the items below because they cannot be recovered from the documentation.”

Refinement: We restructured items 21–27 to focus on organizational practices rather than document accessibility. We changed the stem from “If an external person had access to documents...” to “During the data analysis phase, the project team...,” shifted focus from document recovery to actual team activities, and added the qualifier “whether formally documented or not” to several items. This preserved construct measurement while eliminating documentation requirements.

Learning and Updating (Theory Revision Block)

Expert Feedback: One reviewer recommended expansion: “I think that the learning/updating part is an interesting and important part of the project. I would expand this section more and ask more details about the updating process and outcomes.”

Refinement: We expanded items 28–29 into a six-item battery covering: (1) recognition of disconfirming evidence, (2) revision of project assumptions, (3) adjustment of resource allocation, (4) modification of success criteria, (5) documentation of lessons learned, and (6) dissemination of insights to organizational stakeholders.

A.3.2 Phase 2: Pilot Survey Feedback

The pilot survey ($n = 16$ expert respondents) was conducted on April 11, 2023. Respondents completed the survey and provided structured feedback after each measurement block.

Block 1: Problem Definition and Theory Development

Response Option Mutual Exclusivity Three respondents questioned the forced-choice format:

Respondent 2: “Unclear why I could not tick more than one answer to each question, or create a ranking.”

Respondent 3: “Some multiple choices are not really ‘mutually exclusive’ and I had to focus and use imagination to decide which one to choose.”

Respondent 12: “I felt that some polar choices were not necessarily mutually exclusive and had a hard time sometimes not being just in the middle without really choosing.”

Refinement: We replaced forced-choice formats with Likert scales (1–5) where respondents indicated the extent to which managers engaged in each practice. We reframed items from comparative (A vs. B) to independent assessments (extent of A; extent of B). This allowed respondents to indicate that managers engaged in both practices to varying degrees.

Use of Examples and Scenarios Two respondents reported that scenarios improved comprehension:

Respondent 3: “This was easier, since there is a scenario. Still, some multiple choices are not really ‘mutually exclusive’.”

Respondent 6: “These questions were easier to understand because the artificial scenario supported my thinking.”

Refinement: We introduced brief vignettes before each measurement block describing a generic managerial scenario. This provided respondents with a concrete reference point while maintaining generalizability across contexts.

Question Interdependence Three respondents noted that items appeared sequentially dependent:

Respondent 1: “A couple of questions seemed to be related to the previous one: answers could be led by previous answers.”

Respondent 14: “The second question contains both a plan to increase sales (that corresponds to my first answer) and mapping resources (which is related to the previous question).”

Respondent 15: “answers to Q4 and Q5 depend a lot on what was selected in Q3.”

Refinement: We randomized item order within blocks to minimize response carryover effects, revised double-barreled questions (e.g., splitting “plan to increase sales and map resources” into separate items), and added transitional text between related items to clarify their distinct foci.

Terminology Accessibility One respondent noted terminology barriers:

Respondent 16: “The questions are easier to understand for someone exposed to the line of ‘our framework’ of scientific approach. For example, the word ‘assumptions’—it is difficult to plug in this word in a normal conversation with managers. We might need to find a simpler word or another question prior to this in order to get them to think through assumptions.”

Refinement: We replaced technical terminology with managerial language: “assumptions” → “initial beliefs about how the project would unfold”; “validation” → “testing” or “checking”; “scenario” → “situation” or “set of conditions.” We added brief definitions in parentheses for unavoidable technical terms.

Block 2: Process Characteristics

Scale Midpoint Interpretation Two respondents questioned midpoint meaning:

Respondent 2: “Unclear whether there could be mixing approaches and positioning in the middle of the scale meant a mix or indifference between the two choices.”

Respondent 16: “Similar concerns that I had for first part of the survey, having the centre which implies both are equally important for me is the more feasible scenario.”

Refinement: We added explicit instructions: “The midpoint (3) indicates that you perceive both approaches as equally characteristic of the decision-making process. It does not indicate indifference or inability to choose.”

Implicit vs. Explicit Constructs One respondent requested clarification:

Respondent 12: “the distinction between implicit assumptions and explicit assumptions should probably be explained.”

Refinement: We added operational definitions: “Explicit assumptions are those that were discussed and documented by the team. Implicit assumptions are those that guided decisions but were not openly discussed or written down.”

Question Translation to Practice Two respondents found items too abstract:

Respondent 2: “One question was unclear to me as I could not translate it into practical terms.”

Respondent 4: “Clear but a bit abstract/nuanced at times which might confuse some.”

Refinement: We added concrete examples in parentheses (e.g., “evidence (such as data, expert opinions, or pilot results)”) and replaced theoretical language with action-oriented phrasing (“considered multiple explanations” instead of “evaluated alternative theoretical frameworks”).

Block 3: Outcomes and Learning

Desirability Bias Three respondents identified social desirability concerns:

Respondent 3: “The last question is leading. Even a non-scientific DM would say that he used ‘data’ and revised his plan. If I am a DM and I want a benchmark, there are some questions that have desirability bias in terms of how decisions ‘should’ be made.”

Respondent 15: “there could be some ‘desirability bias’ / ‘overconfidence’ playing in the last answers, where you ask how the decision maker reacts to new pieces of information that challenge the initial plan—it is a bit ‘undesirable’ to answer that, vis-à-vis information that challenges the initial plan, a decision maker won’t take it into account.”

Refinement: We reframed items to ask about observed organizational practices rather than ideal behaviors (“In practice, when evidence challenged expectations, the team typically...” instead of “How should managers respond...”), added neutral response options (“The team proceeded with the original plan while monitoring for additional signals”), introduced behavioral frequency items (“How often did the team...”), and included reverse-coded items to detect acquiescence bias.

Question Clarity and Answer Length Two respondents noted comprehension difficulties:

Respondent 8: “Do you tend to assess the expected outcomes of a plan by focusing on its payoff and objectives OR by comparing it to a benchmark?—this question was not very clear, in theory I could evaluate the payoffs by comparing them to a benchmark?”

Respondent 15: “I found this section a bit difficult to follow. For example, Q4 had very long answers ‘(whether it is favorable and not favorable)’—maybe as a rule of thumb you could have max-1 line long answers.”

Refinement: We limited response options to one line (approximately 10–12 words), separated conceptually distinct elements into separate items (payoff evaluation and benchmark comparison became two items), and removed parenthetical qualifiers from response options (moving explanatory text to item stems).

Repetition Across Blocks Two respondents perceived redundancy:

Respondent 2: “Also some questions sound repetitive of those that I answered at the beginning. Is that because the context is different?”

Respondent 12: “same perception as block 1” [referring to mutual exclusivity concerns].

Refinement: We added transitional text explaining how later blocks assessed implementation rather than planning, ensured distinct verb tenses (planning phase: “would,” implementation phase: “did”), and revised items to focus on distinct behavioral indicators within each process stage.

Respondent Burden Two respondents noted survey length:

Respondent 6: “This survey took a little over 20 minutes to complete.”

Respondent 5: “This block itself was not taxing but since it is the last section it became a bit tiring.”

Refinement: We eliminated eight redundant items identified through pilot data factor analysis, introduced progress indicators between blocks, and shortened the initial project description section. Target completion time: 15 minutes.

A.3.3 Summary of Major Refinements

Table A3 summarizes the major item refinements resulting from expert review and pilot feedback.

Table A3: Summary of Major Item Refinements from Expert Review Process

Issue Identified	Original Formulation	Refined Formulation
Abstract stakeholder reference	“key stakeholders”	“the project team including yourself”
Vague action terminology	“an action by the firm”	“a strategic initiative or operational change”
Unclear scenario concept	“scenario (combination of dimensions)”	“situation defined by specific resources, conditions, and activities”
Documentation assumption	“If external person had access to documents...”	“During the data analysis phase, the team...”
Technical jargon	“assumptions,” “validation,” “baseline reasoning”	“initial beliefs,” “testing,” “initial expectations”
Forced-choice constraints	“A or B” format	Likert scale (1–5) for extent of A and extent of B
Long response options	Parenthetical qualifiers in options	Brief options; qualifiers moved to stems
Desirability bias risk	“How should managers...”	“In practice, the team typically...”

A.3.4 Methodological Lessons

The iterative expert review process yielded five methodological insights.

Construct-Language Gap: Academic constructs (e.g., “scenario,” “validation,” “baseline reasoning”) often lacked clear operational referents for practitioners. We translated theoretical concepts into concrete managerial behaviors and decisions.

False Dichotomies: Initial forced-choice formats presumed mutually exclusive managerial approaches. Pilot feedback revealed that managers employ multiple approaches simultaneously. Likert scales better captured this behavioral complexity.

Context Dependence: Abstract items were difficult for respondents to answer without concrete context. Brief vignettes substantially improved question comprehensibility without sacrificing generalizability.

Social Desirability: Items asking about “ideal” or “should” behaviors were susceptible to socially desirable responding. Reframing items to focus on actual organizational practices reduced this bias.

Respondent Burden: Expert feedback on survey length and cognitive difficulty proved essential. Items rated as “tiring” or “hard to understand” in pilot testing were revised or eliminated.

These lessons informed not only the specific item refinements documented above but also our broader approach to scale development in organizational contexts characterized by normative pressure and behavioral complexity.

A.4 Missing Data Patterns

Table A4: Missingness Patterns Analysis

Panel A: Overall Missingness Summary		
Total Observations	9,468	
Complete Cases	8,028	84.79%
Incomplete Cases	1,440	15.21%
Panel B: Distribution of Missing Variables per Case		
No Missing Variables	8,028	84.79%
1 Missing Variable	904	9.55%
2 Missing Variables	297	3.14%
3 Missing Variables	104	1.10%
4 Missing Variables	51	0.54%
5+ Missing Variables	84	0.89%
Panel C: Temporal Pattern (Survey Position)		
Early Items (SDM 1–8)	1,148	1.52%
Middle Items (SDM 9–16)	716	0.95%
Late Items (SDM 17–24)	1,088	1.44%
Panel D: Missingness by Hierarchical Tier		
Tier 1 (CEO/Top Management)	554	0.44%
Tier 2 (Senior Management)	744	0.41%
Tier 3 (Middle Management)	2,039	0.72%
Tier 4 (Junior Management)	1,877	0.79%
Tier 5 (Frontline)	4,254	1.26%
Panel E: Missingness by SDM Dimension		
Theorizing	1,237	1.31%
Empirical Testing	1,171	1.24%
Updating	544	0.57%
Panel F: Significant Predictors of Missingness		
<i>Positive Associations:</i>		
Decision Area = Other/Missing	$\beta = 1.15$	$p < 0.001$
Education = Missing	$\beta = 3.00$	$p = 0.019$
<i>Negative Associations (vs. smallest firm):</i>		
All Firm Size Categories	$\beta = -1.42$ to -2.14	$p < 0.01$

Notes: Panel A shows overall data completeness. Panel B presents the distribution of cases by number of missing variables; 84.79% of cases have complete data on all 28 SDM items. Panel C examines temporal patterns across survey sections, showing relatively uniform missingness rates (0.95–1.52%). Panel D reveals a hierarchical gradient in missingness, with frontline employees (1.26%) having nearly three times the rate of top management (0.44%). Panel E shows missingness varies by dimension, with Updating showing the lowest rate (0.57%). Panel F reports results from a Poisson regression predicting total missing items per respondent; larger firms consistently show lower missingness. Percentages in Panels C–E represent the proportion of cells with missing data within each category. N = 9,468 respondents across 61 firms.

A.5 Item-Level Correlations

Table A5: Correlation Matrix: Scientific Decision-Making Items

	sdm1	sdm2	sdm3	sdm4	sdm5	sdm6	sdm7	sdm8	sdm9	sdm10	sdm11	sdm12	sdm13	sdm14	sdm15	sdm16	sdm17	sdm18	sdm19	sdm20	sdm21	sdm22	sdm23	sdm24
sdm1	1.00	.21	.09	-.03	-.04	.10	.01	.00	.00	-.01	-.01	.02	-.03	.01	.05	.03	.05	.04	.11	.03	-.04	.00	.11	.06
sdm2	.21	1.00	.02	.00	-.14	.01	-.11	-.04	-.07	.07	-.08	-.08	.08	-.05	.04	.09	-.07	-.03	.05	-.03	-.01	.01	.06	.00
sdm3	.09	.02	1.00	-.05	.02	.02	.00	.01	.10	-.05	.01	.06	-.02	.07	.03	.01	.10	.03	.11	-.01	.04	-.01	.07	.02
sdm4	-.03	.00	-.05	1.00	-.02	.00	.00	.09	.13	.04	.03	.06	-.03	.10	.06	-.08	.02	.06	.00	-.04	.10	.11	.03	.03
sdm5	-.04	-.14	.02	-.02	1.00	.01	.08	-.01	.03	-.08	.06	.06	-.03	.02	-.05	-.04	.03	.02	-.04	.02	.03	-.03	-.04	-.04
sdm6	.10	.01	.02	.00	.01	1.00	.02	-.01	-.03	-.02	.00	-.04	-.03	-.01	.06	.00	.03	.06	.02	.00	-.06	-.03	.02	.04
sdm7	.01	-.11	.00	.00	.08	.02	1.00	.04	.06	-.05	.14	.10	-.10	.03	-.05	-.08	.11	.05	-.08	.33	.00	.00	-.06	.04
sdm8	.00	-.04	.01	.09	-.01	-.01	.04	1.00	.21	.01	.05	.13	-.06	.11	.04	-.06	.12	.11	.09	.06	.12	.08	.07	.05
sdm9	.00	-.07	.10	.13	.03	-.03	.06	.21	1.00	.01	.03	.20	-.15	.16	.06	-.09	.16	.11	.10	.00	.11	.06	.11	.04
sdm10	-.01	.07	-.05	.04	-.08	-.02	-.05	.01	.01	1.00	-.03	-.01	.05	-.02	.03	.04	-.03	-.02	.01	-.04	-.07	.02	.07	.06
sdm11	-.01	-.08	.01	.03	.06	.00	.14	.05	.03	-.03	1.00	.20	-.13	.06	.03	-.07	.01	.11	.04	-.01	.11	.04	.01	.00
sdm12	.02	-.08	.06	.06	.06	-.04	.10	.13	.20	-.01	.20	1.00	-.16	.12	.04	-.08	.13	.07	.08	.04	.07	.04	.08	.05
sdm13	-.03	.08	-.02	-.03	-.03	-.03	-.10	-.06	-.15	.05	-.13	-.16	1.00	-.13	-.09	.15	-.08	-.10	-.04	-.04	.01	-.01	-.08	-.10
sdm14	.01	-.05	.07	.10	.02	-.01	.03	.11	.16	-.02	.06	.12	-.13	1.00	.12	-.11	.09	.07	.06	-.01	.10	.07	.07	.02
sdm15	.05	.04	.03	.06	-.05	.06	-.05	.04	.06	.03	.03	.04	-.09	.12	1.00	-.05	.08	.04	.12	-.02	.04	.05	.15	.11
sdm16	.03	.09	.01	-.08	-.04	.00	-.08	-.06	-.09	.04	-.07	-.08	.15	-.11	-.05	1.00	-.09	-.06	.04	.03	-.05	-.04	-.01	-.04
sdm17	.05	-.07	.10	.02	.03	.03	.11	.12	.16	-.03	.01	.13	-.08	.09	.08	-.09	1.00	.12	.11	.06	.07	.06	.10	.07
sdm18	.04	-.03	.03	.06	.02	.06	.05	.11	.11	-.02	.06	.07	-.10	.07	.04	-.06	.12	1.00	.10	-.03	.07	.02	.12	.10
sdm19	.11	.05	.11	.00	-.04	.02	-.08	.09	.10	.01	-.01	.08	-.04	.06	.12	.04	.11	.10	1.00	-.07	.09	.04	.31	.09
sdm20	.03	-.03	-.01	-.04	.02	.00	.33	.06	.00	-.04	.11	.04	-.04	-.01	-.02	.03	.06	-.03	-.07	1.00	-.03	.02	-.05	.06
sdm21	-.04	-.01	.04	.10	.03	-.06	.00	.12	.11	-.07	.04	.07	.01	.10	.04	-.05	.07	.07	.09	-.03	1.00	.21	.05	.00
sdm22	.00	.01	-.01	.11	-.03	-.03	.00	.08	.06	.02	.01	.04	-.01	.07	.05	-.04	.06	.02	.04	.02	.21	1.00	.06	.07
sdm23	.11	.06	.07	.03	-.04	.02	-.06	.07	.11	.07	.00	.08	-.08	.07	.15	-.01	.10	.12	.31	-.05	.05	.06	1.00	.19
sdm24	.06	.00	.02	.03	-.04	.04	.04	.05	.04	.06	.00	.05	-.10	.02	.11	-.04	.07	.10	.09	.06	.00	.07	.19	1.00

Note. Pearson correlation coefficients between all 24 SDM items. Values represent pairwise correlations based on available data (N ranges from 9,122 to 9,430 depending on missing values). Correlations $\geq |.10|$ are considered meaningful for interpretation. Leading zeros omitted for space efficiency. Diagonal values (1.00) represent perfect self-correlation.

B Psychometric Details

This appendix provides full CFA results and inter-factor correlation matrices supporting Section 4. Tables B.1–B.2 document model fit across CFA specifications and the formative specification test; Tables B.3–B.4 report factor-level and observed correlations among SDM dimensions. Table B5 contrasts CFA-estimated and composite-level interfactor correlations, documenting the modeling artifact discussed in Section 4.5.

Table B1: Confirmatory Factor Analysis: Model Fit Statistics and Comparisons

Model	<i>N</i>	Fit Indices						Model Comparison		
		χ^2	<i>df</i>	CFI	TLI	RMSEA	AIC	$\Delta\chi^2$	Δdf	<i>p</i>
1-Factor	8,028	5,541.20	252	.462	.411	.051	94.62	—	—	—
2-Factor	8,028	5,421.51	251	.474	.422	.051	96.65	119.70	1	< .001
3-Factor	8,028	5,291.37	249	.487	.432	.050	100.68	130.14	2	< .001

Note. All models converged successfully. Model comparison column shows chi-square difference tests versus previous model (nested comparisons). All χ^2 tests and model comparisons significant at $p < .001$. CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; RMSEA = Root Mean Square Error of Approximation; AIC = Akaike Information Criterion. Conventional fit thresholds: CFI/TLI $\geq .90$ (acceptable) or $\geq .95$ (good); RMSEA $\leq .08$ (acceptable) or $\leq .05$ (good). Statistical significance does not necessarily imply practical superiority; theoretical interpretability and parsimony should be considered.

Table B2: Formative vs. Reflective Specification: Model Fit Comparison

Specification	χ^2	<i>df</i>	CFI	TLI	RMSEA	Note
<i>Reflective specifications</i>						
Reflective (single factor)	2342.7	252	0.389	0.330	0.055	
Reflective (2 factors)	2294.4	251	0.403	0.343	0.054	
Reflective (3 factors)	2239.6	249	0.418	0.355	0.054	
Second-order reflective	2247.2	249	0.416	0.352	0.054	
<i>Formative specification</i>						
Formative (composite indicators)	—	—	—	—	—	VIF _{max} = 1.23

Note. Reflective CFAs estimated with *semopy* using listwise deletion ($N = 8,028$ complete cases). Bold CFI values indicate poor fit (CFI < .90). Formative specification is evaluated via VIF (Variance Inflation Factor) rather than CFA fit indices; VIF < 3 confirms non-redundancy among indicators. Items sdm2, sdm10, sdm13, sdm16 are reverse-coded. HO = second-order (hierarchical) reflective CFA with SDM as second-order factor.

Table B3: Factor Correlations in CFA Models

Model	Factor Pair	Covariance	Correlation
2-Factor	Causal Reasoning ↔ Evidence Evaluation	−0.001	−.74
	Causal Reasoning ↔ Evidence Evaluation	−0.001	−.88
3-Factor	Causal Reasoning ↔ Adaptive Learning	−0.001	−.48
	Empirical Testing ↔ Adaptive Learning	0.040	.95

Note. Factor correlations from confirmatory factor analysis models. Two-factor model separates theorizing and empirical assessment. Three-factor model additionally isolates belief updating as a distinct dimension. High correlations between empirical assessment and belief updating (.95) suggest potential collinearity. Negative interfactor correlations involving Causal Reasoning are a modeling artifact of imposing reflective assumptions on formative indicators; composite-score correlations are positive for all pairs. Covariance values are unstandardized; correlations are standardized.

C Experimental Priming Analyses

Table B4: Latent and Observed Correlations Among SDM Dimensions

Panel A. CFA Latent Factor Correlations			
Model	Factor Pair	Covariance	Correlation
2-Factor	Causal Reasoning ↔ Evidence Evaluation	−0.001	−.74
3-Factor	Causal Reasoning ↔ Evidence Evaluation	−0.001	−.88
	Causal Reasoning ↔ Adaptive Learning	−0.001	−.48
	Evidence Evaluation ↔ Adaptive Learning	0.040	.95
Panel B. Observed Composite-Score Correlations			
	(1)	(2)	(3)
(1) Causal Reasoning	—		
(2) Evidence Evaluation	.164***	—	
(3) Adaptive Learning	.117***	.257***	—

Note. Panel A reports standardized latent factor correlations from confirmatory factor analysis (CFA) models. Covariances are unstandardized; correlations are standardized. Negative interfactor correlations involving Causal Reasoning reflect modeling artifacts that can arise when imposing reflective assumptions on conceptually formative indicators. Panel B reports Pearson correlations among observed composite scores ($N = 9,449\text{--}9,456$). All $p < .001$. Unlike the latent CFA correlations, observed composite correlations are positive across all SDM dimensions, supporting nomological coherence of the construct.

Table B5: Factor Intercorrelations: CFA-Estimated vs. Composite-Score Level

	CR × EE	CR × AL	EE × AL
CFA factor correlations	−.88	−.48	.95
Composite-score correlations	.164***	.117***	.257***

Note. CR = Causal Reasoning; EE = Evidence Evaluation; AL = Adaptive Learning. CFA correlations from the three-factor reflective model (Appendix B1; CFI = .487). Composite-score correlations are Pearson r on subscale means ($N \approx 9,450$). Negative CFA values reflect estimation artifacts when a reflective model is imposed on formative indicators; the factor covariance matrix absorbs the misfit, producing implausible correlation estimates (Jarvis et al., 2003; Dolan, 1994). Composite-level correlations represent the substantive association among subdimensions. *** $p < .001$.

Table C.1: Appendix Table A2 — Descriptive Statistics and Pairwise Comparisons by SDM Component

	Mean	SD	<i>N</i>	Cohen's <i>d</i>
<i>Panel A: Overall SDM</i>				
Scientific prime	3.582	0.416	140	—
Placebo	3.071	0.323	183	—
Intuitive prime	2.801	0.327	139	—
<i>Pairwise differences (reference: Placebo)</i>				
Scientific – Placebo	+0.511***			1.396
Placebo – Intuitive	+0.269***			0.829
Scientific – Intuitive	+0.780***			2.086
ANOVA: $F(2, 459) = 175.21, p < .001$				
<i>Panel B: Causal Reasoning</i>				
Scientific prime	3.540	0.508	140	—
Placebo	2.916	0.384	183	—
Intuitive prime	2.726	0.431	139	—
<i>Pairwise differences (reference: Placebo)</i>				
Scientific – Placebo	+0.624***			1.413
Placebo – Intuitive	+0.191***			0.471
Scientific – Intuitive	+0.815***			1.730
ANOVA: $F(2, 459) = 134.11, p < .001$				
<i>Panel C: Evidence Evaluation</i>				
Scientific prime	3.496	0.483	140	—
Placebo	3.139	0.438	183	—
Intuitive prime	2.760	0.504	139	—
<i>Pairwise differences (reference: Placebo)</i>				
Scientific – Placebo	+0.357***			0.779
Placebo – Intuitive	+0.379***			0.811
Scientific – Intuitive	+0.736***			1.492
ANOVA: $F(2, 459) = 84.77, p < .001$				
<i>Panel D: Adaptive Learning</i>				
Scientific prime	3.886	0.736	140	—
Placebo	3.325	0.667	183	—
Intuitive prime	3.094	0.704	139	—
<i>Pairwise differences (reference: Placebo)</i>				
Scientific – Placebo	+0.562***			0.805
Placebo – Intuitive	+0.231***			0.338
Scientific – Intuitive	+0.793***			1.100
ANOVA: $F(2, 459) = 47.89, p < .001$				

Note. Mean differences are in SDM-point units (scale: 1–5). Cohen's *d* computed using the pooled SD of the two conditions being compared. All pairwise differences significant at $p < .001$ (Tukey HSD). *** $p < .001$.

Table C.2: Appendix Table A3 — Item Means by Condition

Item	Step	Sci	Placebo	Intu	SD	N
<i>Causal Reasoning</i>						
sdm1	Problem Diagnosis	3.371	2.333	2.273	1.299	462
sdm2	Problem Diagnosis	3.158	2.383	2.180	1.201	461
sdm3	Causal Theorizing	3.671	2.514	2.691	1.355	462
sdm5	Causal Theorizing	3.264	2.753	2.559	1.255	458
sdm6	Problem Diagnosis	4.129	2.747	2.827	1.349	461
sdm8	Causal Theorizing	3.813	3.617	3.182	1.205	456
sdm9	Causal Theorizing	4.330	3.936	3.594	1.091	287
sdm10	Hypothesis Testing	3.657	3.366	2.676	1.367	462
sdm11	Hypothesis Testing	3.479	2.956	2.388	1.278	462
sdm12	Hypothesis Testing	4.064	3.099	3.029	1.292	458
sdm13	Hypothesis Testing	2.236	2.880	2.899	1.308	462
<i>Evidence Evaluation</i>						
sdm4	Evidence Gathering	4.321	4.251	3.604	1.032	462
sdm7	Evidence Appraisal	3.071	2.650	2.108	1.328	462
sdm14	Evidence Gathering	3.950	3.473	2.748	1.234	461
sdm15	Evidence Gathering	3.630	3.473	2.583	1.275	459
sdm16	Evidence Appraisal	2.243	2.421	2.856	1.270	462
sdm17	Evidence Appraisal	3.721	3.341	2.978	1.258	461
sdm18	Evidence Appraisal	3.796	3.140	2.717	1.298	454
sdm19	Evidence Appraisal	3.593	2.863	2.920	1.377	460
sdm20	Evidence Appraisal	3.158	2.652	2.317	1.273	459
<i>Adaptive Learning</i>						
sdm21	Belief Adjustment	3.714	3.508	3.604	1.257	462
sdm22	Belief Adjustment	3.900	3.720	3.432	1.100	461
sdm23	Belief Adjustment	3.900	3.077	3.187	1.257	462
sdm24	Belief Adjustment	4.029	3.000	2.151	1.322	461

Note. Item means (scale: 1–5) by experimental condition. Sci = Scientific prime, Intu = Intuitive prime. SD = pooled standard deviation across all three conditions. Items sorted by subscale and step. *** $p < .001$, ** $p < .01$, * $p < .05$.

Scientific Treatment Text

THE SCIENTIFIC APPROACH TO IMPORTANT (MANAGERIAL) DECISIONS

The scientific method, pioneered by Galileo Galilei, Nicholas Copernicus, and Sir Francis Bacon, led to an explosion of knowledge for humanity.

While the scientific method had remained largely unused by the business community, managers are starting to adopt a “scientific approach” to their decision making. This approach entails:

1. Identifying the right problem

Think deeply about the strategic problem that needs to be addressed. Finding the ‘right’ problem to solve is more important than solving the ‘wrong’ problem.

2. Using logic to outline multiple scenarios and plans of action

Consider alternative formulations of the problem and alternative plans to solve it. Think deeply and gather information to address questions such as:

- Do I even need a decision in the first place?
- What are the alternative plans that I can consider?
- What are my assumptions for each plan? Are they explicit and transparent to all key stakeholders?
- What are the future conditions that I find most likely? Am I focusing on them?
- How would each of the alternative plans address my problem?

3. Formulating hypotheses

Break down the problem and your plans to identify precise elements that need ex ante validation for the plan to work as expected. For instance:

- Assumptions that need validation.
- Specific cause-and-effect relationships that need to be true.

4. Designing, conducting, and analyzing “experiments”

Collect information to test your plans. However, a scientific decision-maker does not test to verify if a plan works, but rather stress-test plans to challenge them, picking the ones that “survive” the testing. Some principles are:

- Define assessment criteria before testing.
- Define a clear benchmark to compare plans against.
- Try to disprove/challenge the hypotheses you created.
- Be “neutral” to plans and do not favor options without testing.
- Final assessments should compare plans to each other (i.e., do not look only at the payoff of a single plan!).

5. Drawing a conclusion

“Scientific managers” draw conclusions flexibly, revising their thinking considering the new information obtained. They will:

- Change their thinking based on new evidence instead of endlessly seeking for data that confirms their thinking.
- Use the tests to decide whether to change the original plan, choose another plan, or proceed with the preferred plan.

EXAMPLE OF THE SCIENTIFIC APPROACH TO IMPORTANT (MANAGERIAL) DECISIONS

Researchers at Bocconi University (an important business school in Europe) found that entrepreneurs who were trained in the scientific approach were significantly more likely to succeed.

One exemplary application is Mimoto, a motorbike sharing service in Milan, Italy.

Table C.3: Experimental Manipulation Effects on SDM

	(1) Overall SDM	(2) Causal Reasoning	(3) Evidence Evaluation	(4) Adaptive Learning
<i>Panel A: Prime only</i>				
Scientific prime	0.511*** (0.043)	0.624*** (0.052)	0.357*** (0.052)	0.562*** (0.080)
Intuitive prime	-0.269*** (0.037)	-0.191*** (0.046)	-0.379*** (0.054)	-0.231** (0.078)
Controls	—	—	—	—
R^2	0.433	0.369	0.270	0.173
N	462	462	462	462
<i>Panel B: Prime + Tier</i>				
Scientific prime	0.536*** (0.044)	0.669*** (0.054)	0.361*** (0.054)	0.579*** (0.083)
Intuitive prime	-0.244*** (0.038)	-0.143** (0.047)	-0.375*** (0.056)	-0.217** (0.080)
Controls	Tier	Tier	Tier	Tier
R^2	0.441	0.392	0.270	0.177
N	462	462	462	462
<i>Panel C: Prime + Tier + Education</i>				
Scientific prime	0.588*** (0.047)	0.719*** (0.055)	0.416*** (0.061)	0.626*** (0.091)
Intuitive prime	-0.235*** (0.042)	-0.141** (0.051)	-0.346*** (0.064)	-0.239** (0.091)
Controls	Tier, Education	Tier, Education	Tier, Education	Tier, Education
R^2	0.511	0.464	0.307	0.229
N	354	354	354	354
<i>Panel D: Prime + Tier + Education + Tenure</i>				
Scientific prime	0.590*** (0.047)	0.724*** (0.055)	0.415*** (0.062)	0.626*** (0.092)
Intuitive prime	-0.236*** (0.042)	-0.145** (0.050)	-0.345*** (0.063)	-0.239** (0.091)
Controls	Tier, Education, Tenure	Tier, Education, Tenure	Tier, Education, Tenure	Tier, Education, Tenure
R^2	0.512	0.469	0.307	0.229
N	354	354	354	354
<i>Panel E: Prime + Tier + Education + Tenure + Decision</i>				
Scientific prime	0.592*** (0.047)	0.731*** (0.056)	0.416*** (0.062)	0.614*** (0.095)
Intuitive prime	-0.234*** (0.043)	-0.148** (0.053)	-0.334*** (0.064)	-0.242** (0.093)
Controls	Tier, Education, Tenure, Decision	Tier, Education, Tenure, Decision	Tier, Education, Tenure, Decision	Tier, Education, Tenure, Decision
R^2	0.519	0.478	0.321	0.236
N	354	354	354	354

Note. OLS with HC3 robust standard errors in parentheses. Reference condition: Placebo. N decreases from 462 to 354 in Panels C–E due to missing covariate data. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table C.4: Appendix Table A2 — Item-Level Priming Effects

Item	Sci $\hat{\beta}$	Intu $\hat{\beta}$
<i>Causal Reasoning (sdm1–sdm13)</i>		
sdm1	1.038*** (0.145)	−0.060 (0.124)
sdm2	0.776*** (0.138)	−0.203 (0.115)
sdm3	1.158*** (0.143)	0.177 (0.141)
sdm4	0.070 (0.098)	−0.647*** (0.122)
sdm5	0.512*** (0.142)	−0.194 (0.134)
sdm6	1.381*** (0.130)	0.080 (0.141)
sdm7	0.421** (0.154)	−0.542*** (0.132)
sdm8	0.196 (0.131)	−0.434** (0.136)
sdm9	0.394** (0.136)	−0.342* (0.162)
sdm10	0.291 (0.150)	−0.690*** (0.146)
sdm11	0.522*** (0.139)	−0.568*** (0.133)
sdm12	0.965*** (0.130)	−0.070 (0.145)
sdm13	−0.644*** (0.143)	0.019 (0.144)
Item	Sci $\hat{\beta}$	Intu $\hat{\beta}$
<i>Evidence Evaluation (sdm14–sdm20)</i>		
sdm14	0.477*** (0.123)	−0.724*** (0.135)
sdm15	0.158 (0.136)	−0.890*** (0.134)
sdm16	−0.178 (0.142)	0.435** (0.138)
sdm17	0.381** (0.138)	−0.362** (0.139)
sdm18	0.656*** (0.138)	−0.422** (0.141)
sdm19	0.729*** (0.151)	0.056 (0.152)
sdm20	0.506*** (0.144)	−0.335* (0.134)
<i>Adaptive Learning (sdm21–sdm24)</i>		
sdm21	0.206 (0.145)	0.096 (0.137)
sdm22	0.180 (0.113)	−0.288* (0.130)
sdm23	0.823*** (0.135)	0.111 (0.138)
sdm24	1.029*** (0.118)	−0.849*** (0.131)

Note. Each row reports coefficients from a separate OLS regression (HC3 robust SEs) of the item score on condition dummies, with Placebo as reference ($N = 462$). Sci = Scientific prime, Intu = Intuitive prime. Standard errors in parentheses. The sole non-significant item (sdm21) shows no response to either prime. *** $p < .001$, ** $p < .01$, * $p < .05$.

Mimoto had imagined many ways in which they could cater to the local market for individual transportation. Through surveys, Mimoto learned that one of the demographics they considered — working women — were more likely to use the service than any other demographic.

However, women didn't like the motorbikes that Mimoto had chosen because they were too bulky and heavy. Based on these results, Mimoto better selected its target market and its product and retested, leading to resounding success.

Implementing the scientific method can be a game changer for innovation, product development, or making decisions that are hard-to-reverse and very impactful to the future of organizations.

Placebo Text

The NoTreatment (placebo) group did not receive any learning/treatment text. Participants in this condition were taken directly to the survey phase without any prior reading phase about a decision-making approach.

Intuitive Treatment Text

THE USE OF INTUITION TO IMPORTANT (MANAGERIAL) DECISIONS

A 2022 Harvard Business Review article offered the following key takeaways about the value of intuition in decision-making:

1. **Recognize the value of intuition**

Intuition, or “gut feelings,” can be an important source of insight and guidance. It can help us make quick decisions, especially when there is limited information available.

2. **Understand the risks of overthinking**

Overthinking can lead to decision paralysis, as well as anxiety and stress. It can also cause us to miss out on opportunities or ignore important information.

3. **Develop self-awareness**

To trust our intuition, we need to be self-aware and in tune with our thoughts and feelings. We should pay attention to our physical sensations, such as a tightening in our stomach or a racing heart, which can indicate when something is off.

4. **Practice mindfulness**

Mindfulness can help us quiet our minds and tune in to our intuition. We can do this by taking deep breaths, focusing on the present moment, and avoiding distractions.

5. **Practice the use of your intuition**

We can build trust in our intuition by making a small decision based on our gut feeling, and then evaluate how it turned out.

6. **Communicate openly and honestly**

We should communicate openly and honestly with teams about the importance of intuition and the risks of overthinking. This creates a culture that encourages intuitive thinking instead of instilling “fear” about sharing one's inner thoughts.

EXAMPLE OF THE USE OF INTUITION IN IMPORTANT (MANAGERIAL) DECISIONS

Let's say that a company is preparing to launch a new product, a high-end electric car. The CEO has been working with a team of engineers and market researchers for months to perfect the design, price point, and marketing strategy for the product. However, despite all the research and testing, the CEO has a gut feeling that something is not quite right.

During a meeting with the team, the CEO expresses his concerns and proposes a last-minute change to the design of the car. The team is initially resistant, as they have already invested significant time and resources into the current design. However, the CEO argues that the change will make the car more appealing to a wider audience and give it a competitive edge in the market.

Despite the lack of time for additional testing, the CEO's intuition wins out, and the team scrambles to implement the new design before the product launch. When the car is finally unveiled, it receives rave reviews and quickly becomes a top seller in its category. The CEO's intuition was proven to be right, and the company's success is attributed to his willingness to trust his gut and make a bold decision.

This example shows how, in some cases, intuition can be more important than theorizing and testing, especially when the stakes are high and time is short. While data and analysis are important, they are not always enough to make the best decisions. CEOs and other leaders who are in tune with their intuition and willing to take risks based on their instincts can sometimes achieve greater success than those who rely solely on data and analysis.

Attention Checks To verify that participants read and processed the priming material, all three conditions included a forced-choice attention check immediately following the prime (Q191), asking participants to identify the decision-making approach described in the passage. Correct-response rates were 99.2% in the Scientific condition, 99.2% in the Intuitive condition, and 72.1% in the Placebo condition. The lower rate in the Placebo condition is consistent with the design: the neutral passage did not advocate a distinct approach, making the 'correct' answer less salient by construction. Participants who failed both attention checks across the survey were excluded prior to analysis; all participants retained in the analytic sample (N=462) passed at least one check.

D ICC Robustness Checks

D.1 ICC Across Model Specifications

Table D1.1 reports ICC estimates using three estimation methods (ANOVA, REML, Conditional REML) for all four SDM outcomes. Estimates are consistent across methods, ranging from 0.5% to 2.0% between-firm variance.

Table D1.1: Intraclass Correlation Estimates by Model Specification

Outcome	Model	ICC	$\sigma^2_{\text{between}}$	σ^2_{within}
SDM Overall	ANOVA	0.0094	0.0008	0.0843
	Mixed Effects (REML)	0.0110	0.0009	0.0843
	Conditional (REML)	0.0126	0.0011	0.0838
Causal Reasoning	ANOVA	0.0065	0.0009	0.1377
	Mixed Effects (REML)	0.0068	0.0009	0.1379
	Conditional (REML)	0.0124	0.0017	0.1376
Evidence Evaluation	ANOVA	0.0151	0.0027	0.1786
	Mixed Effects (REML)	0.0200	0.0036	0.1786
	Conditional (REML)	0.0184	0.0033	0.1786
Adaptive Learning	ANOVA	0.0077	0.0029	0.3705
	Mixed Effects (REML)	0.0074	0.0028	0.3702
	Conditional (REML)	0.0054	0.0020	0.3674

Note. This table compares three methods for estimating intraclass correlation coefficients using the full dataset ($N = 9,468$, $K = 61$ firms). ANOVA provides unconditional estimates, Mixed Effects (REML) uses restricted maximum likelihood, and Conditional (REML) adjusts for individual-level covariates.

D.2 ICC Across Subsamples

Table D1.2 extends the comparison to multiple subsamples: minimum firm size restrictions ($n \geq 10$, $n \geq 20$), tier-balanced subsamples, and firm-size-focused subsamples (top 8 only; excluding largest firm; excluding top 8). All ICCs remain below 5%.

D.3 Leave-One-Firm-Out (LOFO) Sensitivity

Table D1.3 reports the range of ICC estimates obtained by removing each of the 61 firms in turn. The maximum perturbation across all outcomes and exclusions is $|\Delta| = 0.003$, confirming that no single firm drives the between-firm clustering estimate. Figure D1.1 plots the distribution of ΔICC across firms for each SDM outcome.

D.4 Balanced-per-Firm Bootstrap

Table D1.4 reports bootstrap ICC estimates under balanced sampling ($m = 20, 30, 40$ observations per firm, 1,000 iterations). Bootstrap estimates are consistent with the full-sample REML estimates.

Table D1.2: Comprehensive ICC Estimates Across Samples and Model Specifications

Sample	Model	ICC by Outcome				K
		Overall	Causal	Evidence	Adaptive	
Full dataset	ANOVA	0.0094	0.0065	0.0151	0.0077	61
	Mixed	0.0110	0.0068	0.0200	0.0074	
	Cond.	0.0126	0.0124	0.0184	0.0054	
$n \geq 10$	ANOVA	0.0099	0.0051	0.0153	0.0082	49
	Mixed	0.0116	0.0056	0.0203	0.0075	
	Cond.	0.0128	0.0107	0.0189	0.0054	
$n \geq 20$	ANOVA	0.0087	0.0043	0.0141	0.0079	38
	Mixed	0.0101	0.0050	0.0177	0.0074	
	Cond.	0.0120	0.0101	0.0161	0.0052	
Balanced higher-ordered tiers	ANOVA	0.0100	0.0044	0.0146	0.0093	31
	Mixed	0.0122	0.0048	0.0196	0.0086	
	Cond.	0.0146	0.0109	0.0178	0.0058	
Balanced 5-tiers	ANOVA	0.0093	0.0038	0.0132	0.0094	21
	Mixed	0.0109	0.0039	0.0163	0.0097	
	Cond.	0.0125	0.0099	0.0152	0.0063	
Top 8 only	ANOVA	0.0092	0.0023	0.0138	0.0096	8
	Mixed	0.0104	0.0025	0.0185	0.0100	
	Cond.	0.0160	0.0084	0.0182	0.0063	
Excl. largest	ANOVA	0.0116	0.0081	0.0184	0.0068	60
	Mixed	0.0124	0.0084	0.0217	0.0072	
	Cond.	0.0134	0.0139	0.0196	0.0044	
Excl. top 8	ANOVA	0.0119	0.0206	0.0214	0.0044	53
	Mixed	0.0128	0.0183	0.0219	0.0067	
	Cond.	0.0137	0.0162	0.0241	0.0018	

Note. This table presents ICC estimates for all combinations of sample restrictions and model specifications. K = number of firms. ANOVA = one-way ANOVA, Mixed = Mixed Effects (REML), Cond. = Conditional (REML) with covariates. Sample restrictions: $n \geq 10$ and $n \geq 20$ retain firms with minimum observations; Bal. higher tiers requires ≥ 3 respondents in tier 1, tiers 2–3, and tiers 4–5; Bal. all tiers requires ≥ 3 in each tier; Top 8 only and Excl. largest/top 8 manipulate firm size distribution.

Table D1.3: ICC Stability: Leave-One-Firm-Out (LOFO) Sensitivity

Outcome	Baseline		LOFO Range		
	ICC	Firms	Min ICC	Max ICC	Max $ \Delta $
SDM Overall	0.011	61	0.009	0.012	0.002
Causal Reasoning	0.007	61	0.005	0.008	0.002
Evidence Evaluation	0.020	61	0.017	0.022	0.003
Adaptive Learning	0.007	61	0.005	0.008	0.002

Note. Leave-one-firm-out (LOFO) analysis: each of the 61 firms is removed in turn and the ICC is re-estimated on the remaining sample. Baseline ICC = REML estimate on the full dataset ($N = 9,458$). LOFO Range shows the minimum and maximum ICC observed across 61 exclusions. Max $|\Delta|$ = maximum absolute change from baseline. No single firm drives the ICC estimate; the maximum perturbation is 0.003, far below the 5% conventional threshold.

Table D1.4: Balanced-per-Firm Bootstrap ICC Results

Outcome	m	ICC Mean	ICC Median	ICC SD	[2.5%, 97.5%]	Δ Mean
Overall	20	0.0079	0.0049	0.0093	[0.0000, 0.0314]	-0.0716
Overall	30	0.0136	0.0128	0.0094	[0.0000, 0.0335]	-0.0659
Overall	40	0.0136	0.0133	0.0077	[0.0000, 0.0295]	-0.0659
Causal Reasoning	20	0.0120	0.0103	0.0108	[0.0000, 0.0376]	-0.0442
Causal Reasoning	30	0.0143	0.0135	0.0098	[0.0000, 0.0369]	-0.0419
Causal Reasoning	40	0.0133	0.0132	0.0075	[0.0000, 0.0294]	-0.0429
Evidence Evaluation	20	0.0171	0.0156	0.0127	[0.0000, 0.0453]	-0.1050
Evidence Evaluation	30	0.0179	0.0173	0.0103	[0.0000, 0.0400]	-0.1042
Evidence Evaluation	40	0.0192	0.0186	0.0084	[0.0048, 0.0374]	-0.1028
Adaptive Learning	20	0.0070	0.0033	0.0088	[0.0000, 0.0306]	-0.0585
Adaptive Learning	30	0.0095	0.0082	0.0084	[0.0000, 0.0292]	-0.0560
Adaptive Learning	40	0.0076	0.0063	0.0065	[0.0000, 0.0228]	-0.0579

Note. Bootstrap ICC estimates with balanced samples per firm (m = observations per firm drawn per bootstrap iteration, 1,000 iterations). Δ Mean = $ICC_{\text{bootstrap}} - ICC_{\text{full sample}}$ (ANOVA). All bootstrap ICCs remain below 5%, confirming robustness to sampling imbalance across firms.

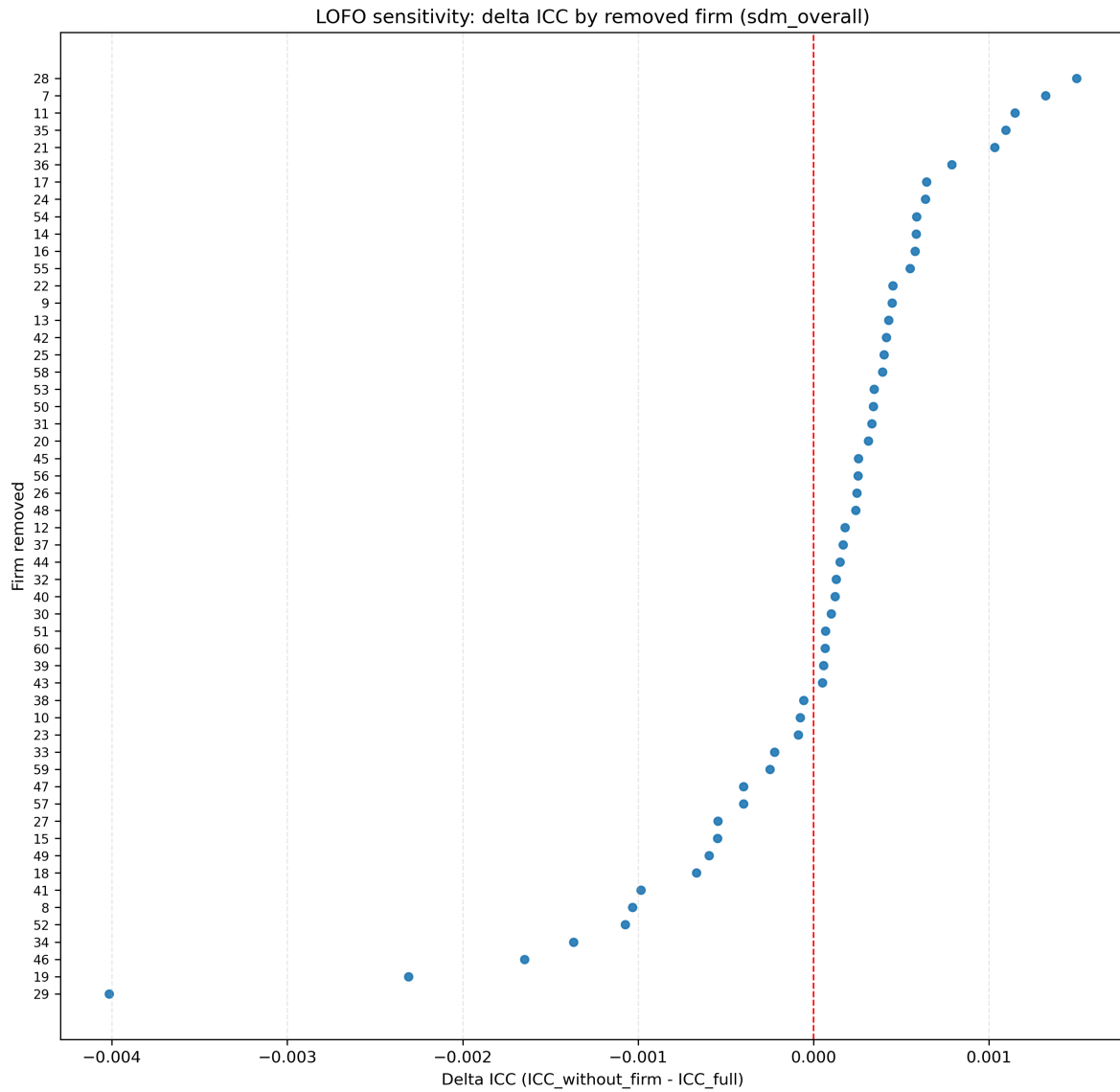


Figure D1.1: LOFO Δ ICC by Firm: Change in ICC When Each Firm is Removed

E Organizational Firm-Level Analyses (Exploratory)

E.1 Descriptive Statistics

Table E.1 reports means, standard deviations, and ranges for all variables used in Section 5.

Table E.1: Descriptive Statistics: Application Sample Variables

Variable	<i>N</i>	Mean	<i>SD</i>	Min	Max
<i>SDM Scales</i>					
SDM Overall	9,458	3.211	0.292	1.000	4.636
Causal Reasoning	9,452	3.047	0.372	1.000	5.000
Evidence Evaluation	9,456	3.343	0.426	1.333	5.000
Adaptive Learning	9,449	3.498	0.611	1.000	5.000
<i>Outcome</i>					
Vignette accuracy	9,468	0.485	0.204	0.000	1.000
<i>Controls</i>					
Hierarchical level	9,468	3.901	1.223	1.000	5.000
Tenure (years)	9,468	11.09	8.699	0.000	30.00
Graduate degree	9,468	0.585	0.493	0.000	1.000
STEM background	9,468	0.276	0.447	0.000	1.000

Note. SDM scales scored 1–5 (item means). Vignette accuracy is the proportion of correct responses across decision scenarios. Hierarchical level coded 1 (CEO/Top) to 5 (Frontline). Graduate degree and STEM background are binary (0/1). Tenure capped at 30 years per survey response format. *N* varies across SDM scales due to item-level missing data.

E.2 Firm-Level Regressions

Table E.2: Hierarchical Level Effects: Three-Tier Specification

	Overall		Causal		Evidence		Adaptive	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Model A: Tiers Only</i>								
Senior (Tiers 1–2)	0.049*** (0.013)	0.061*** (0.018)	0.034* (0.017)	0.060** (0.022)	0.013 (0.019)	0.018 (0.021)	0.123*** (0.027)	0.163*** (0.029)
Middle (Tiers 3–4)	0.016* (0.007)	0.029** (0.010)	0.023** (0.009)	0.047*** (0.010)	-0.023* (0.010)	-0.006 (0.016)	0.044** (0.014)	0.053* (0.022)
<i>Model B: + Education</i>								
Senior (Tiers 1–2)	0.043*** (0.013)	0.055** (0.018)	0.032 (0.017)	0.056* (0.022)	0.008 (0.019)	0.014 (0.021)	0.118*** (0.027)	0.155*** (0.029)
Middle (Tiers 3–4)	0.011 (0.007)	0.027** (0.009)	0.021* (0.009)	0.046*** (0.010)	-0.026** (0.010)	-0.007 (0.016)	0.039** (0.014)	0.050* (0.021)
<i>Model C: + Decision Type</i>								
Senior (Tiers 1–2)	0.049*** (0.013)	0.059** (0.018)	0.037* (0.017)	0.058* (0.023)	0.017 (0.019)	0.020 (0.020)	0.129*** (0.028)	0.161*** (0.031)
Middle (Tiers 3–4)	0.015* (0.007)	0.030** (0.010)	0.023** (0.009)	0.047*** (0.011)	-0.019 (0.010)	-0.002 (0.016)	0.045** (0.014)	0.052* (0.021)
Firm Fixed Effects	No	Yes	No	Yes	No	Yes	No	Yes
Education Controls				Added in Model B				
Decision Type Controls				Added in Model C				
Observations	9,458		9,452		9,456		9,449	

Note. Each column presents results for a different outcome variable. Odd-numbered columns report OLS estimates; even-numbered columns report firm fixed effects estimates with standard errors clustered at the firm level. Standard errors in parentheses. Reference category is Tier 5 (Frontline). Senior = Tiers 1–2 combined, Middle = Tiers 3–4 combined. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

F Scoring Robustness: Minimum Operator

This appendix addresses the question raised by the formative necessity interpretation: if each subdimension of SDM is a non-substitutable component of the construct, should the weakest subdimension—rather than the average—define the composite score? We compare the conventional mean score with a minimum operator, $\min(\text{CR}, \text{EE}, \text{AL})$, across the three key analyses: experimental sensitivity, vignette prediction, and the hierarchy gradient.

Table F.1 reports the RCT comparison. All condition means are on the 1–5 response scale for comparability with Table 4. Panel A shows that the scientific prime produces large, significant effects under both scoring methods ($d \geq 1.11$ for scientific vs. placebo; $\eta^2 \geq .355$), with effect sizes approximately 14% smaller under the minimum operator—an expected result, since the minimum is bounded by the weakest dimension and therefore more variable. Panel B shows that the prime elevated all three subdimensions simultaneously, ruling out the possibility that mean-based gains mask a near-zero component: all three per-dimension d values exceed the conventional large-effect threshold ($d > 0.78$).

Table F.2 reports the field-sample comparison. Vignette prediction is nearly identical under both scoring methods ($r = .175$ vs. $r = .166$; firm-FE coefficients of .120 and .092, both $p < .001$). The hierarchy gradient is, if anything, steeper under the minimum operator ($b = -.021$ vs. $b = -.017$), indicating that the most conservative scoring method strengthens rather than attenuates the main empirical finding.

Panel C documents the distribution properties of each scoring method and identifies which subdimension most frequently serves as the binding constraint. Causal Reasoning is the weakest dimension for 61.5% of managers, meaning the minimum score captures primarily CR performance. The two scoring methods correlate at $r = .784$, confirming they capture the same rank-ordering of managers while differing in their distributional properties. Conclusions are unchanged across all key analyses; we retain the mean score as the primary estimator on grounds of statistical stability and comparability with conventional scale-composite practice.

Table F.1: Experimental Sensitivity: Average vs. Minimum-Operator Scoring

<i>Panel A: Overall SDM score</i>							
Scoring	Condition Means			Cohen's d		$F(2, 459)$	η^2
	Scientific ($n = 140$)	Placebo ($n = 183$)	Intuitive ($n = 139$)	Sci-Plc	Sci-Int		
Average	3.58	3.07	2.80	1.40	2.09	175.21***	.433
Minimum	3.22	2.73	2.41	1.11	1.81	126.44***	.355

Panel B: Per-dimension condition means and prime effect (Sci vs. Placebo)

Dimension	M Scientific	M Placebo	M Intuitive	d (Sci-Plc)
Causal Reasoning	3.54	2.92	2.73	1.41
Evidence Evaluation	3.50	3.14	2.76	0.78
Adaptive Learning	3.89	3.33	3.09	0.80

Note. *** $p < .001$. All means are on the 1–5 response scale. *Minimum* = $\min(\text{CR}, \text{EE}, \text{AL})$ per respondent. The per-dimension results in Panel B confirm that scientific priming elevated all three subdimensions simultaneously, ruling out the possibility that average-based gains mask a near-zero contribution from one component.

Table F.2: Field-Sample Robustness: Average vs. Minimum-Operator Scoring ($N = 9,458$)

<i>Panel A: Vignette prediction</i>					
Scoring	Pearson r	b (firm FE)	SE	p	N
Average	.175***	.120	.007	< .001	9,458
Minimum	.166***	.092	.006	< .001	9,458

<i>Panel B: Hierarchy gradient (OLS with firm fixed effects)</i>							
Scoring	$b(\text{tiers})$	SE	Tier mean SDM score				
			T1 ($n = 554$)	T2 ($n = 744$)	T3 ($n = 2,038$)	T4 ($n = 1,877$)	T5 ($n = 4,245$)
Average	−.017***	.003	3.25	3.22	3.22	3.21	3.20
Minimum	−.021***	.004	2.93	2.90	2.90	2.90	2.87

<i>Panel C: Score distribution and binding dimension</i>				
Scoring	M	SD	% below midpoint (2.5)	r with average
Average	3.21	0.29	0.6%	1.00
Minimum	2.89	0.36	9.6%	.784***

Dimension most often binding (= minimum)	% of managers
Causal Reasoning	61.5%
Evidence Evaluation	21.8%
Adaptive Learning	20.4%

Note. *** $p < .001$. *Minimum* = min(CR, EE, AL) per respondent, computed from the three subdimension mean scores on the 1–5 response scale. $b(\text{tiers})$ is the OLS coefficient from regressing SDM score on tier (1 = top, 5 = frontline) with firm fixed effects; a negative coefficient indicates senior managers score higher. Panel C reports which subdimension is the binding constraint (i.e., equals the minimum) for each manager. Source: 24b_scoring_sensitivity_min.py.